

2^k Factorial plan

Chapter 6

- **Special Case** of Factorial Designs with k factors and **two** levels per each factor
- Usually the two levels are named **Low** and **High** (as usual, the factors can be quantitative or qualitative, in the ANOVA context the factor metric is useless)
- These kind of designs are used in “**Factor screening experiments**”
- They could be “*building blocks*” of other experimentation techniques. (i.e. RSM)
- The analysis methods are the **same** we discussed, with a **special language**
- Let us introduce the **standard labels** (1), a, b, ab

Statistical Model (two factors)

$$y_{ijk} = \mu + \tau_i + \beta_j + \tau\beta_{ij} + \varepsilon$$

$$i \in \{-, +\} \quad j \in \{-, +\}$$

$$i \in \{-1, +1\} \quad j \in \{-1, +1\}$$

$$\sum_i \hat{\tau}_i = 0 \Rightarrow \hat{\tau}_+ + \hat{\tau}_- = 0 \Rightarrow \hat{\tau}_+ = -\hat{\tau}_-$$

$$\sum_j \hat{\beta}_j = 0 \Rightarrow \hat{\beta}_+ + \hat{\beta}_- = 0 \Rightarrow \hat{\beta}_+ = -\hat{\beta}_-$$

$$\sum_i \widehat{\tau\beta}_{ij} = 0 \Rightarrow \begin{array}{l} \widehat{\tau\beta}_{++} + \widehat{\tau\beta}_{-+} = 0 \\ \widehat{\tau\beta}_{+-} + \widehat{\tau\beta}_{--} = 0 \end{array} \Rightarrow \begin{array}{l} \widehat{\tau\beta}_{++} = -\widehat{\tau\beta}_{-+} \\ \widehat{\tau\beta}_{+-} = -\widehat{\tau\beta}_{--} \end{array}$$

$$\sum_j \widehat{\tau\beta}_{ij} = 0 \Rightarrow \begin{array}{l} \widehat{\tau\beta}_{++} + \widehat{\tau\beta}_{+-} = 0 \\ \widehat{\tau\beta}_{-+} + \widehat{\tau\beta}_{--} = 0 \end{array} \Rightarrow \begin{array}{l} \widehat{\tau\beta}_{++} = -\widehat{\tau\beta}_{+-} \\ \widehat{\tau\beta}_{-+} = -\widehat{\tau\beta}_{--} \end{array}$$

$$\widehat{\tau\beta}_{++} = \widehat{\tau\beta}_{--} = -\widehat{\tau\beta}_{-+} = -\widehat{\tau\beta}_{+-}$$

Remark: four information pieces -> four model parameters

Parameter estimates

Let's use **known formulas** and **standard labels**

$$\hat{\mu} = \bar{y}_{...} = \frac{1}{4n} [(1) + a + b + ab]$$

$$\hat{\tau}_+ = \bar{y}_{+..} - \bar{y}_{...} = \frac{a + ab}{2n} - \frac{1}{4n} [(1) + a + b + ab] = \frac{-(1) + a - b + ab}{4n}$$

$$\hat{\tau}_+ = -\hat{\tau}_-$$

$$\hat{\beta}_+ = \bar{y}_{.+.} - \bar{y}_{...} = \frac{b + ab}{2n} - \frac{1}{4n} [(1) + a + b + ab] = \frac{-(1) - a + b + ab}{4n}$$

$$\hat{\beta}_+ = -\hat{\beta}_-$$

$$\widehat{\tau\beta}_{--} = \bar{y}_{--.} - \bar{y}_{-..} - \bar{y}_{.+.} + \bar{y}_{...} = \frac{(1) - a - b + ab}{4n}$$

$$\widehat{\tau\beta}_{--} = \widehat{\tau\beta}_{++} = -\widehat{\tau\beta}_{+-} = -\widehat{\tau\beta}_{-+}$$

Estimate of Factor Effects

(Main Effects and Interactions)

$$\widehat{EFF}_A = \hat{\tau}_+ - \hat{\tau}_- = 2\hat{\tau}_+ = \frac{1}{2n}[ab + a - b - (1)]$$

$$\widehat{EFF}_B = \hat{\beta}_+ - \hat{\beta}_- = 2\hat{\beta}_+ = \frac{1}{2n}[ab + b - a - (1)]$$

$$\widehat{EFF}_{AB} = \frac{\hat{\tau}\hat{\beta}_{++} + \hat{\tau}\hat{\beta}_{--}}{2} - \frac{\hat{\tau}\hat{\beta}_{+-} + \hat{\tau}\hat{\beta}_{-+}}{2} = 2\hat{\tau}\hat{\beta}_{++} = \frac{1}{2n}[ab + (1) - b - a]$$

$$2\hat{\tau}_+ = \widehat{EFF}_A = \frac{\widehat{CONTRAST}_A}{2n}$$

$$2\hat{\beta}_+ = \widehat{EFF}_B = \frac{\widehat{CONTRAST}_B}{2n}$$

$$2\hat{\tau}\hat{\beta}_{++} = \widehat{EFF}_{AB} = \frac{\widehat{CONTRAST}_{AB}}{2n}$$

Note that the contrasts could be negative

Splitting the Sum of Squares

(for Orthogonal designs)

$$SS_{TOT} = SS_A + SS_B + SS_{AB} + SS_E$$

$$SS_A = bn \sum_i \hat{\tau}_i^2 = 2n \sum_{i=-,+} \hat{\tau}_i^2 = n4\hat{\tau}_+^2 = n(2\hat{\tau}_+)^2 = nEFF_A^2 = n \left(\frac{CONTRAST_A}{2n} \right)^2 =$$

$$= n \left(\frac{\overline{CONTRAST_A}}{2n} \right)^2 = \frac{\overline{CONTRAST_A}^2}{4n}$$

$$SS_A = nEFF_A^2 = \frac{\overline{CONTRAST_A}^2}{4n}$$

The same formula structure applies to the other SSs

Design matrix

Std Order	Total	A	B
1	(1)	-	-
2	a	+	-
3	b	-	+
4	ab	+	+

$$\widehat{EFF}_B = \frac{-(1) - a + b + ab}{2n}$$

$$\widehat{EFF}_A = \frac{-(1) + a - b + ab}{2n}$$

How can we find the contrast coefficient?

We build the **Sign Table** (also called **Model Matrix**) by adding columns to the **Design Matrix**

	I	A	B	AB
1 (1)	+	—	—	+
2 a	+	+	—	—
3 b	+	—	+	—
4 ab	+	+	+	+

$$\widehat{EFF}_{[]} = \frac{\widehat{CONTRAST}_{[]}}{2n}$$

$$SS_{[]} = \frac{\widehat{CONTRAST}_{[]}^2}{4n} = n\widehat{EFF}_{[]}^2$$

1) The product of two columns is meaningful

2) $A * A = A^2 = I$ $B * B = B^2 = I$ $I * A = A * I = A$

$I * B = B * I = B$ $A * B = AB$

3) $A^T B = 0$

Residual computation

$$e_{ijk} = y_{ijk} - \hat{y}_{ij}$$

Studentized
residuals

$$e_{ijk}^* = \frac{e_{ijk}}{\hat{V}(e_{ijk})}$$

$$\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \tau\hat{\beta}_{ij}$$

$$\hat{\tau}_- = -\widehat{EFF}_A / 2 \quad \hat{\tau}_{A_+} = \widehat{EFF}_A / 2$$

$$\hat{\beta}_- = -\widehat{EFF}_B / 2 \quad \hat{\beta}_{B_+} = \widehat{EFF}_B / 2$$

$$\tau\hat{\beta}_{--} = \tau\hat{\beta}_{++} = \widehat{EFF}_{AB} / 2$$

$$\tau\hat{\beta}_{-+} = \tau\hat{\beta}_{+-} = -\widehat{EFF}_{AB} / 2$$

Chemical process example

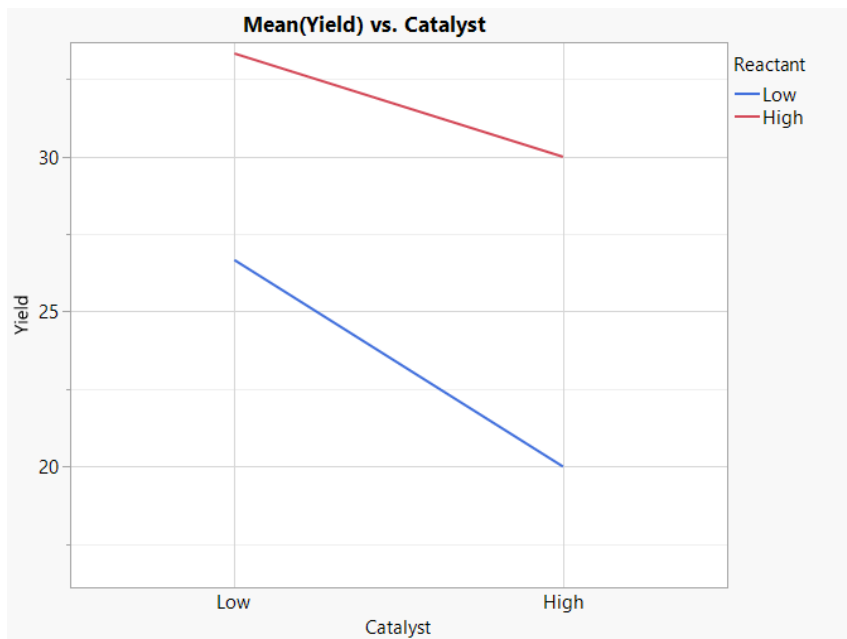
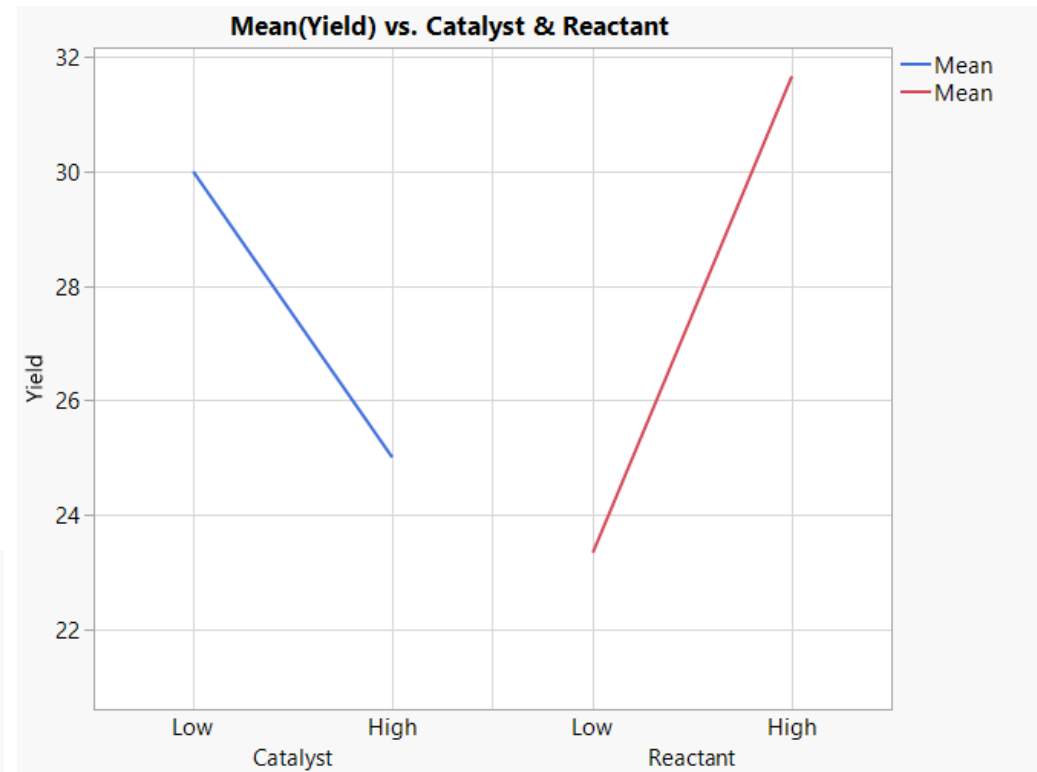
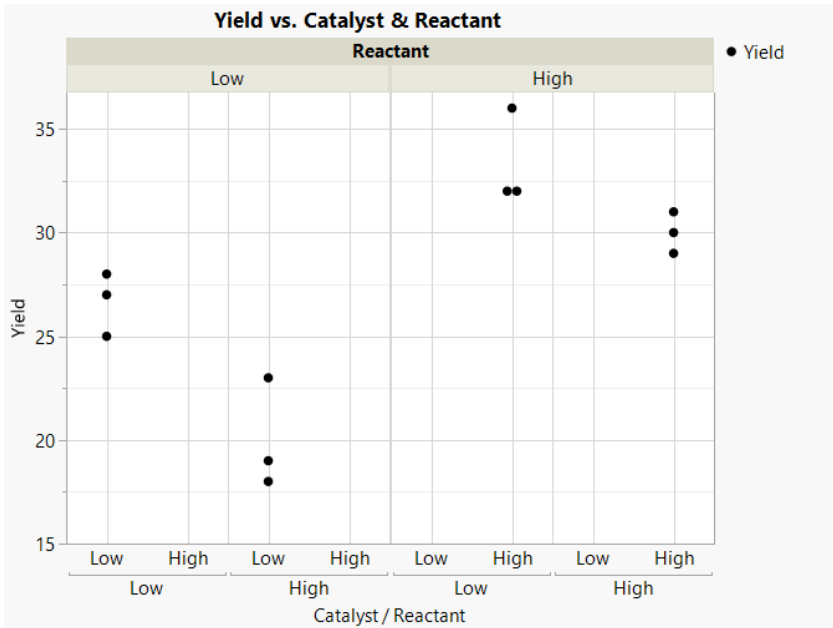
Factor		Treatment Combination	Replicate			Total
<i>A</i>	<i>B</i>		I	II	III	
–	–	<i>A</i> low, <i>B</i> low	28	25	27	80
+	–	<i>A</i> high, <i>B</i> low	36	32	32	100
–	+	<i>A</i> low, <i>B</i> high	18	19	23	60
+	+	<i>A</i> high, <i>B</i> high	31	30	29	90

A = Concentration of the reactant,

B = Amount of catalyst,

y = Process yield

Initial plots



Computations for 6.1 example

		I	A	B	AB	
1	(1)	+	−	−	+	80
2	a	+	+	−	−	100
3	b	+	−	+	−	60
4	ab	+	+	+	+	90

$$n=3$$

$$k=2$$

$$\hat{\mu} = \frac{80 + 100 + 60 + 90}{3 \cdot 4} = 27.5$$

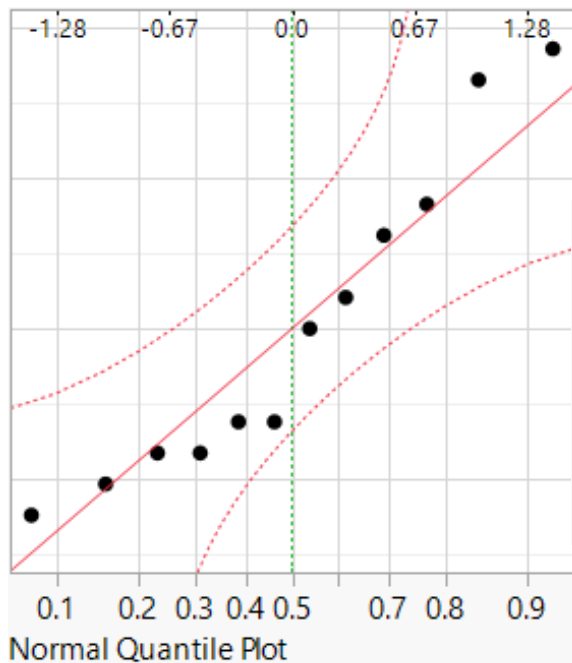
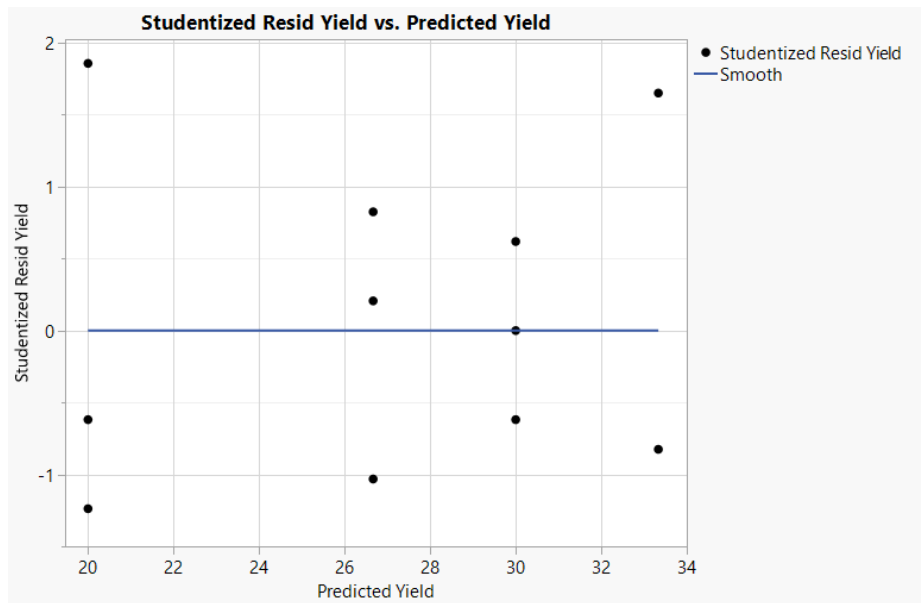
$$\widehat{EFF}_A = \frac{(-80 + 100 - 60 + 90)}{3 \cdot 2} = \frac{50}{6} = 8.33 \quad \widehat{EFF}_B = \frac{(-80 - 100 + 60 + 90)}{6} = -\frac{30}{6} = -5.00$$

$$\widehat{EFF}_{AB} = \frac{(80 - 100 - 60 + 90)}{6} = \frac{10}{6} = 1.67$$

$$SS_A = \frac{50^2}{3 \cdot 2^2} = 208.33 \quad SS_B = 75.00 \quad SS_{AB} = 8.33$$

$$SS_{TOT} = \sum_{i,j,k} y_{ijk}^2 - \frac{y_{...}^2}{2^k n} = 323.0 \quad SS_E = \text{by subtraction}$$

Studentized Residual analysis (full model)

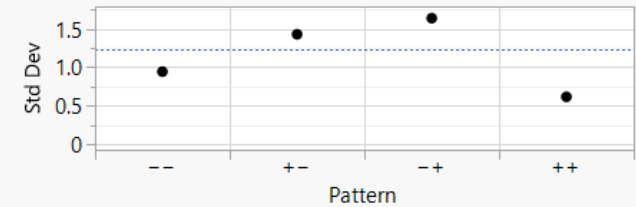


Goodness-of-Fit Test

	W	Prob < W
Shapiro-Wilk	0.9059993	0.1895
	A ²	Simulated p-Value
Anderson-Darling	0.4596571	0.2168

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .

Tests that the Variances are Equal



Level	Count	Std Dev	MeanAbsDif to Mean	MeanAbsDif to Median
--	3	0.945313	0.687614	0.618853
+-	3	1.429179	1.100183	0.825137
-+	3	1.637330	1.237705	1.031421
++	3	0.618853	0.412568	0.412568

Test	F Ratio	DFNum	DFDen	Prob > F
O'Brien[.5]	0.6365	3	8	0.6122
Brown-Forsythe	0.2020	3	8	0.8921
Levene	1.8667	3	8	0.2136
Bartlett	0.5483	3	.	0.6493

Warning: Small sample sizes. Use Caution.

Welch's Test

Welch Anova testing Means Equal, allowing Std Devs Not Equal

F Ratio	DFNum	DFDen	Prob > F
0.0000	3	4.1657	1.0000

ANOVA Table

Summary of Fit

RSquare	0.902993
RSquare Adj	0.866615
Root Mean Square Error	1.979057
Mean of Response	27.5
Observations (or Sum Wgts)	12

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	291.66667	97.2222	24.8227
Error	8	31.33333	3.9167	Prob > F
C. Total	11	323.00000		0.0002*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Reactant	1	1	208.33333	53.1915	<.0001*
Catalyst	1	1	75.00000	19.1489	0.0024*
Reactant*Catalyst	1	1	8.33333	2.1277	0.1828

Conclusions

More information

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	27.5	0.571305	48.14	<.0001*	.
Reactant[Low]	-4.166667	0.571305	-7.29	<.0001*	1
Catalyst[Low]	2.5	0.571305	4.38	0.0024*	1
Reactant[Low]*Catalyst[Low]	0.8333333	0.571305	1.46	0.1828	1

A->Reactant

B->Catalyst

$$\hat{\tau}_{-} = -4.167$$

$$\hat{\tau}_{+} + \hat{\tau}_{-} = 0 \rightarrow \hat{\tau}_{+} = -\hat{\tau}_{-} = 4.167$$

$$\widehat{EFF}_A = \hat{\tau}_{+} - \hat{\tau}_{-} = -2\hat{\tau}_{-} = 8.334$$

ANOVA and Linear Regression

$$y_{ijk} = \mu + \tau_i + \beta_j + \tau\beta_{ij} + \varepsilon$$

$$y = c_0 + c_A x_A + c_B x_B + c_{AB} x_A x_B + \varepsilon$$

$$x_A \in \{-1, +1\} \quad x_B \in \{-1, +1\}$$



$$\hat{c}_0 = \hat{\mu} \quad \hat{c}_A = \frac{\widehat{EFF}_A}{2} = \hat{\tau}_+ \quad \hat{c}_B = \frac{\widehat{EFF}_B}{2} = \hat{\beta}_+ \quad \hat{c}_{AB} = \frac{\widehat{EFF}_{AB}}{2} = \hat{\tau\beta}_{++}$$

$$\hat{\sigma}^2 = MS_E$$

i.e.

$$\begin{aligned} \hat{y}(A_- B_+) &= c_0 + \frac{\widehat{EFF}_A}{2}(-1) + \frac{\widehat{EFF}_B}{2}(+1) + \frac{\widehat{EFF}_{AB}}{2}(-1)(+1) = \\ &= \hat{\mu} + \hat{\tau}_- + \hat{\beta}_+ + \hat{\tau\beta}_{-+} \end{aligned}$$

How could we use the regression?

The regression function can be used to predict the response variable for treatment values for which we didn't make the experiment (but only if the Factor is/are quantitative and inside the experimentation region or slightly outside)

For the chemical process the real factor values were coded to [-1,1]

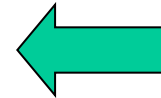
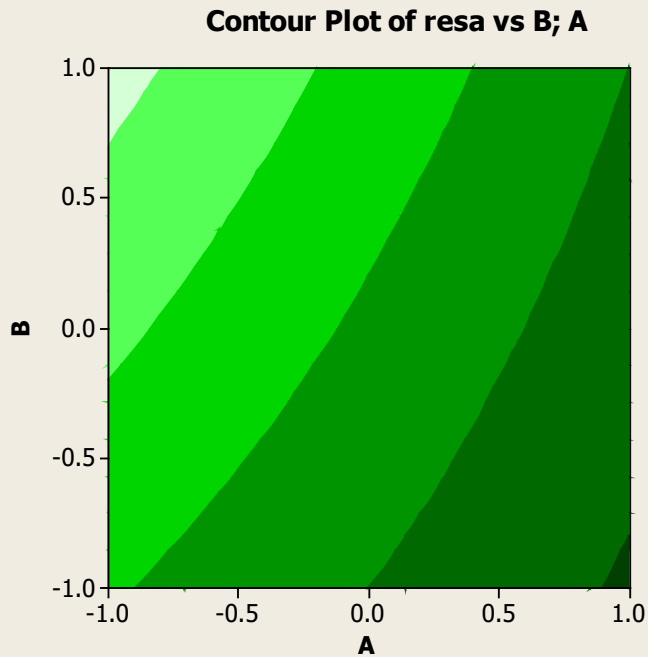
Big advantage: if you code, the coefficients of the regression equation will be numerically **comparable**.

$$x_A = \frac{Conc - (Conc_{High} + Conc_{Low}) / 2}{(Conc_{High} - Conc_{Low}) / 2} \quad x_B = \frac{Cat - (Cat_{High} + Cat_{Low}) / 2}{(Cat_{High} - Cat_{Low}) / 2}$$

$$Conc = \{15\%, 25\% \} \quad Cat = \{1\text{pound}, 2\text{pounds}\}$$

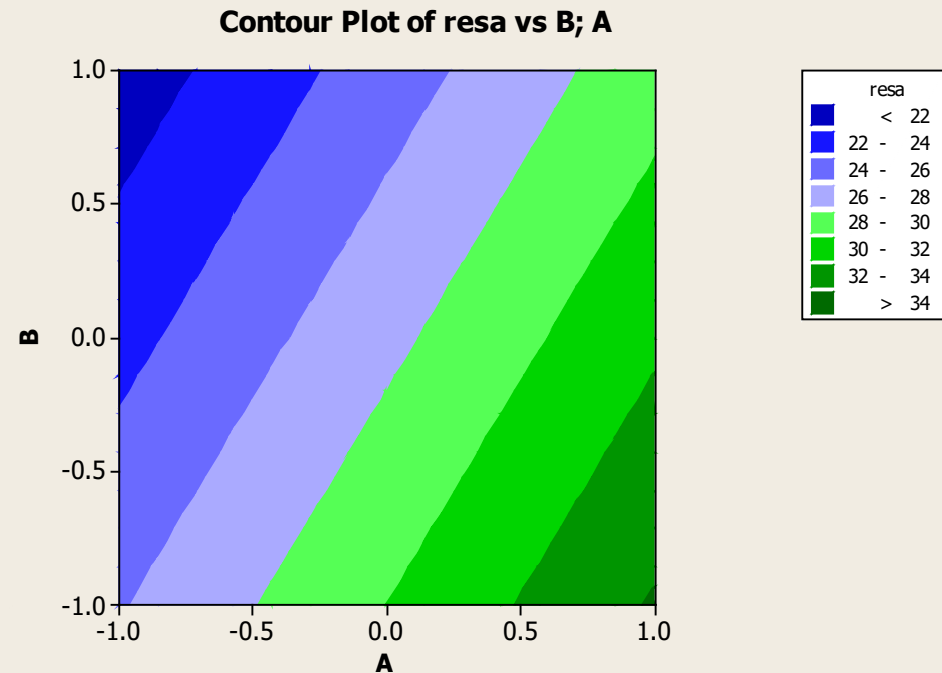
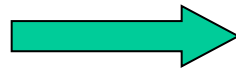
$$\hat{y} = 27.5 + 4.167x_A - 2.5x_B + 0.833x_Ax_B$$

The response surface (coded values)

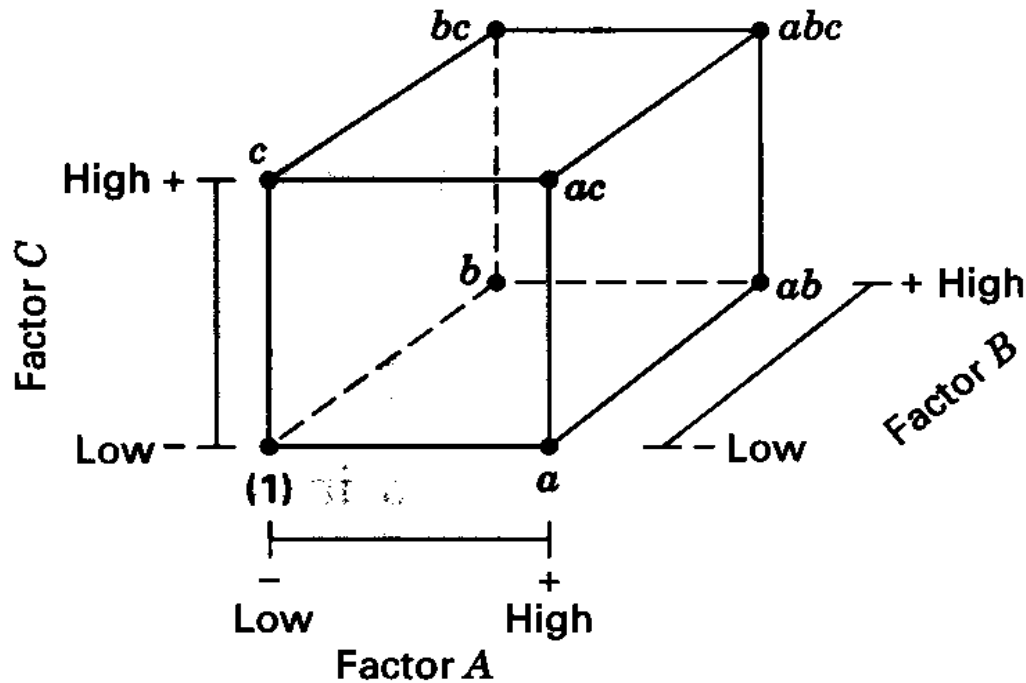


Full model

Reduced
Model



The 2^3 design



(a) Geometric view

Std Order	Factor		
	A	B	C
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-
5	-	-	+
6	+	-	+
7	-	+	+
8	+	+	+

(b) The design matrix

Figure 6-4 The 2^3 factorial design.

Model Matrix (Sign Table) 2^3

		I	A	B	C	AB	BC	AC	ABC
1	(1)	+	—	—	—	+	+	+	—
2	a	+	+	—	—	—	+	—	+
3	b	+	—	+	—	—	—	+	+
4	ab	+	+	+	—	+	—	—	—
5	c	+	—	—	+	+	—	—	+
6	ac	+	+	—	+	—	—	+	—
7	bc	+	—	+	+	—	+	—	—
8	abc	+	+	+	+	+	+	+	+

AB Contrast signs

In general for For 2^k Designs

$$\widehat{EFF}_{[]} = \frac{\widehat{CONTRAST}_{[]}}{n2^{k-1}}$$

$$\hat{y} = \hat{c}_0 + \hat{c}_A x_A + \hat{c}_B x_B + \hat{c}_C x_C + \hat{c}_{AB} x_A x_B + \hat{c}_{AC} x_A x_C \\ + \hat{c}_{BC} x_B x_C + \hat{c}_{ABC} x_A x_B x_C$$

$$\hat{c}_{[]} = \frac{\widehat{EFF}_{[]}}{2}$$

$$SS_{[]} = \frac{\widehat{CONTRAST}_{[]}^2}{n2^k} = n2^{k-2} \widehat{EFF}_{[]}^2 = n2^k \hat{c}_{[]}^2$$

Probability Distribution of Model Parameters

- Probability Distribution of the **Effects** $\widehat{EFF}_{[]} = \frac{\widehat{CONTRAST}_{[]}}{n2^{k-1}}$

$\widehat{CONTRAST}_{[]} \rightarrow$ has 2^k sums of n observations each

$$V\left(\widehat{CONTRAST}_{[]}\right) = 2^k n \sigma^2$$

$$\left\{ \begin{array}{l} \widehat{EFF}_{[]} \sim \text{Normal} \\ E\left(\widehat{EFF}_{[]}\right) = EFF_{[]} \\ V\left(\widehat{EFF}_{[]}\right) = \frac{1}{n2^{k-2}} \sigma^2 \rightarrow \frac{1}{n2^{k-2}} MS_E \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \frac{\widehat{EFF}_{[]} - EFF_{[]}}{\sqrt{\frac{1}{n2^{k-2}} \sigma^2}} \sim N(0,1) \\ \frac{\widehat{EFF}_{[]} - EFF_{[]}}{\sqrt{\frac{1}{n2^{k-2}} MS_E}} \sim t(df_E) \end{array} \right.$$

Probability distribution of the Model Coefficient

$$\hat{c}_{[]} = \frac{\widehat{EFF}_{[]}}{2}$$

$$V(\hat{c}_{[]}) = V\left(\frac{\widehat{EFF}_{[]}}{2}\right) = \frac{1}{4} \frac{\sigma^2}{n2^{k-2}} = \frac{\sigma^2}{n2^k} \rightarrow \hat{V}(\hat{c}_{[]}) = \frac{MS_E}{n2^k}$$

$$se(\hat{c}_{[]}) = \sqrt{V(\hat{c}_{[]})} = \sqrt{\frac{\sigma^2}{n2^k}} \text{ or } \sqrt{\frac{MS_E}{n2^k}}$$

Confidence Interval (without the Bonferroni approach)

$$\hat{c}_{[]} - t_{\alpha/2}(df_E)se(\hat{c}_{[]}) \leq c_{[]} \leq \hat{c}_{[]} + t_{\alpha/2}(df_E)se(\hat{c}_{[]})$$

2³ Example

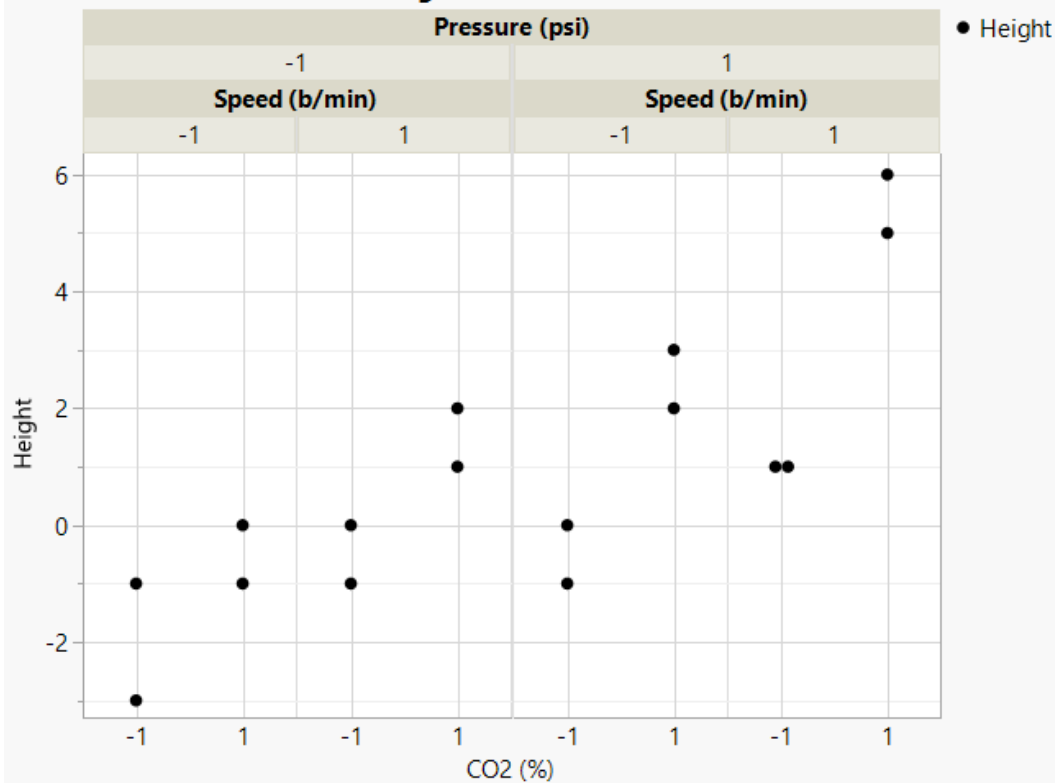
Let's consider again the experiment about the bottling process and suppose the CO₂ percentages are 10% and 12%

Observed data:

	A	B	C	r1	r2		LOW	HIGH
1	-	-	-	-3	-1			
2	+	-	-	0	-1	A (%)	10	12
3	-	+	-	-1	0	B (psi)	25	30
4	+	+	-	2	3	C(b/min)	200	250
5	-	-	+	-1	0			
6	+	-	+	2	1			
7	-	+	+	1	1			
8	+	+	+	6	5			

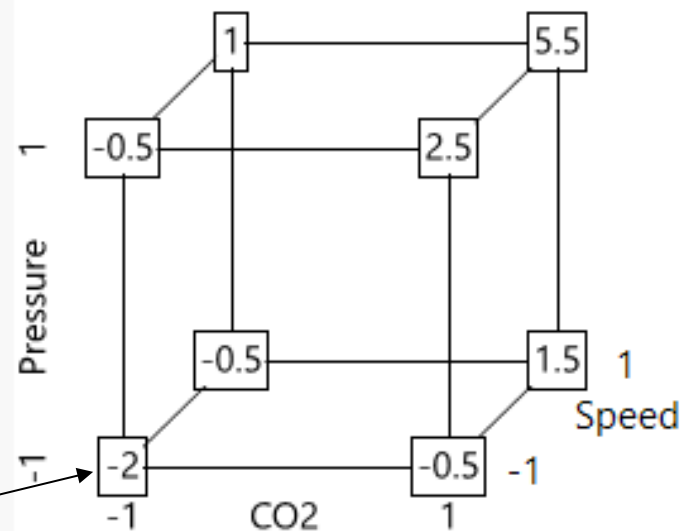
A = %CO₂, **B** = pressure, **C** = Line speed, **y** = Fill height –Target

Height vs. CO2

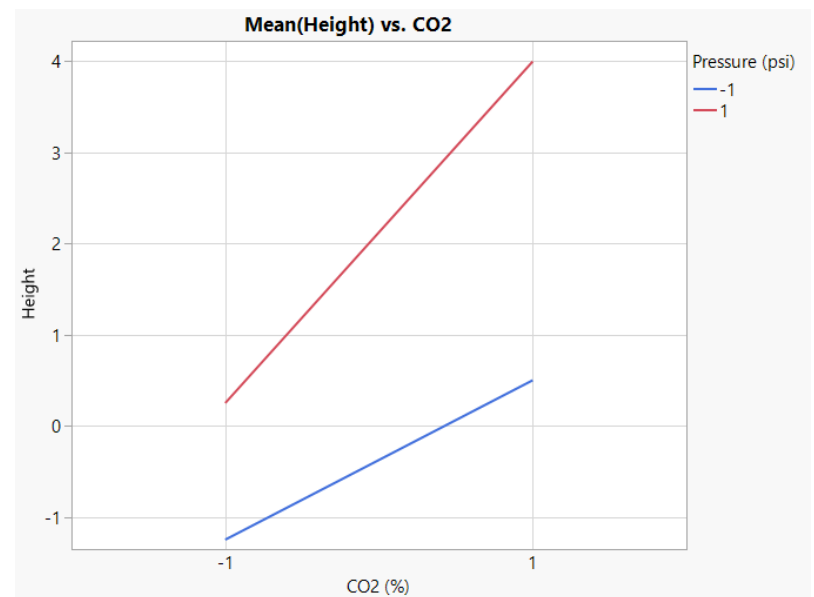
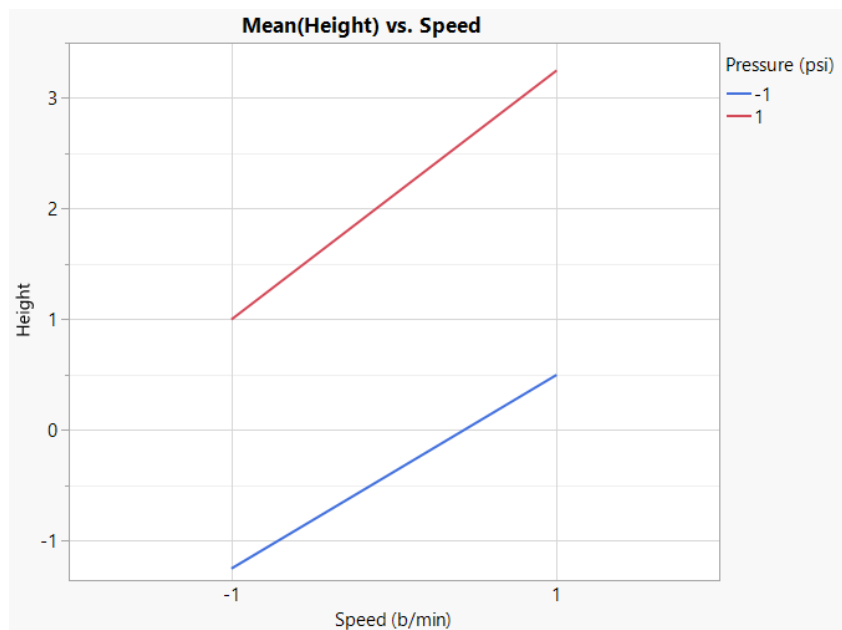
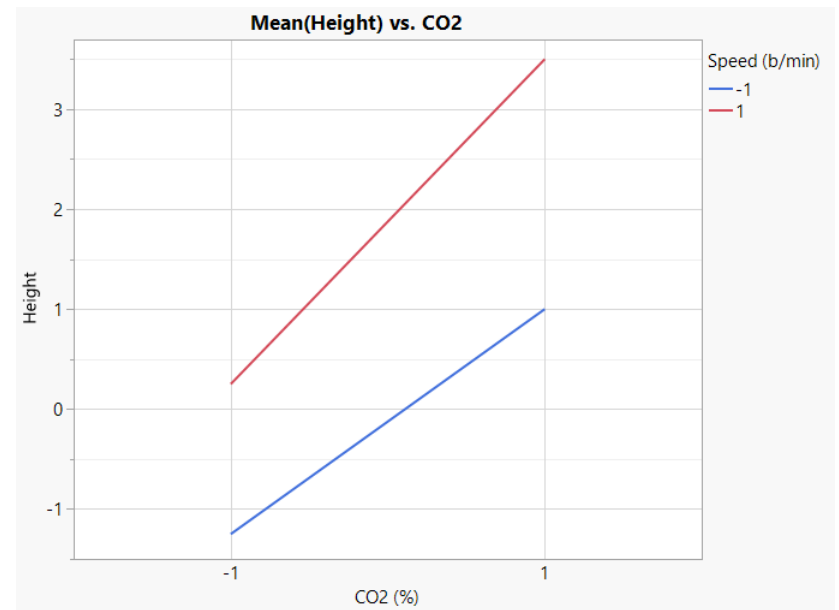
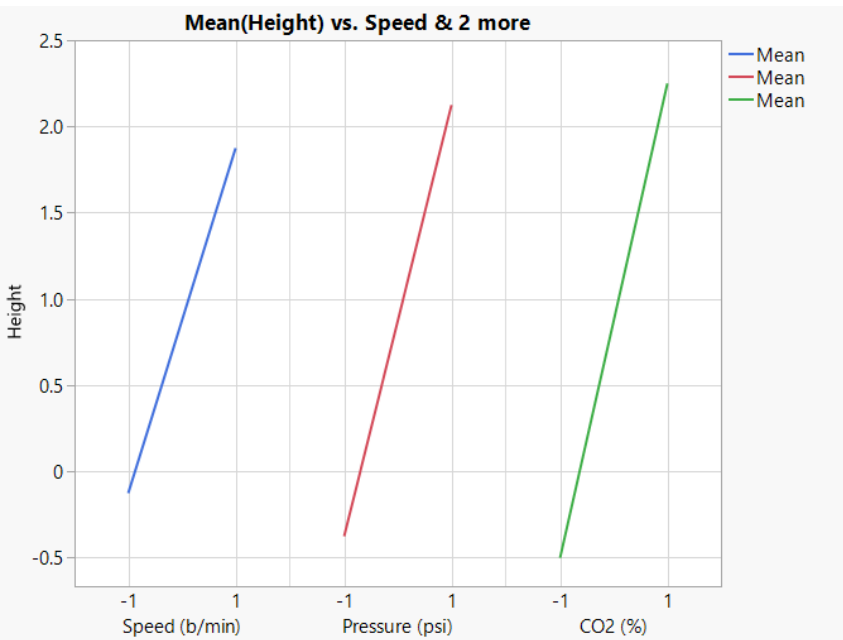


Data Plots

Cube Plot



Mean value



Sign Table for 2³ Design

n=2

		Effect						
	<i>I</i>	<i>A</i>	<i>B</i>	<i>AB</i>	<i>C</i>	<i>AC</i>	<i>BC</i>	<i>ABC</i>
(1) = -4	+	-	-	+	-	+	+	-
<i>a</i> = -1	+	+	-	-	-	-	+	+
<i>b</i> = -1	+	-	+	-	-	+	-	+
<i>ab</i> = 5	+	+	+	+	-	-	-	-
<i>c</i> = -1	+	-	-	+	+	-	-	+
<i>ac</i> = 3	+	+	-	-	+	+	-	-
<i>bc</i> = 2	+	-	+	-	+	-	+	-
<i>abc</i> = 11	+	+	+	+	+	+	+	+
Contr.		22	20	8	16	4	2	2
Effect		2.75	2.50	1.0	2.0	0.5	0.25	0.25

ANOVA Table – Full Model

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	7	76.750000	10.9643	17.5429
Error	8	5.000000	0.6250	Prob > F
C. Total	15	81.750000		0.0003*

Summary of Fit

RSquare	0.938838
RSquare Adj	0.885321
Root Mean Square Error	0.790569
Mean of Response	0.875
Observations (or Sum Wgts)	16

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
CO2	1	1	30.250000	48.4000	0.0001*
Pressure	1	1	25.000000	40.0000	0.0002*
Speed	1	1	16.000000	25.6000	0.0010*
CO2*Pressure	1	1	4.000000	6.4000	0.0353*
CO2*Speed	1	1	1.000000	1.6000	0.2415
Pressure*Speed	1	1	0.250000	0.4000	0.5447
CO2*Pressure*Speed	1	1	0.250000	0.4000	0.5447

Parameter Estimates

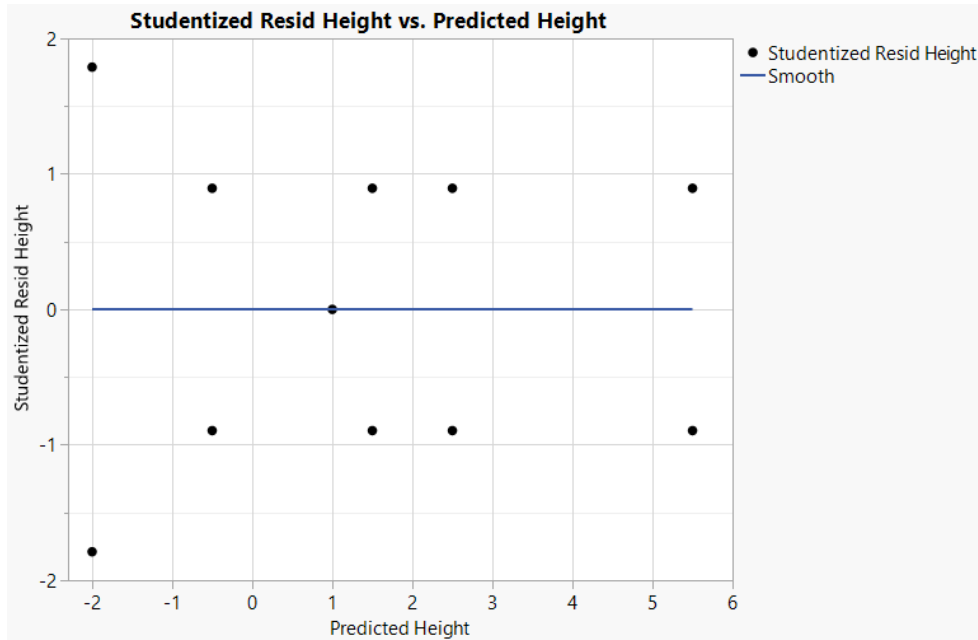
Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	0.875	0.197642	4.43	0.0022*	.
CO2[-1]	-1.375	0.197642	-6.96	0.0001*	1
Pressure[-1]	-1.25	0.197642	-6.32	0.0002*	1
Speed[-1]	-1	0.197642	-5.06	0.0010*	1
CO2[-1]*Pressure[-1]	0.5	0.197642	2.53	0.0353*	1
CO2[-1]*Speed[-1]	0.25	0.197642	1.26	0.2415	1
Pressure[-1]*Speed[-1]	0.125	0.197642	0.63	0.5447	1
CO2[-1]*Pressure[-1]*Speed[-1]	-0.125	0.197642	-0.63	0.5447	1

$$se(\mathbf{CO}_2) = \sqrt{MS_E / (n2^k)} = \sqrt{0.625 / (2 * 2^3)} = 0.198$$

Note: $F_{CO_2} = t_{CO_2}^2$. WHY?

Residual Analysis

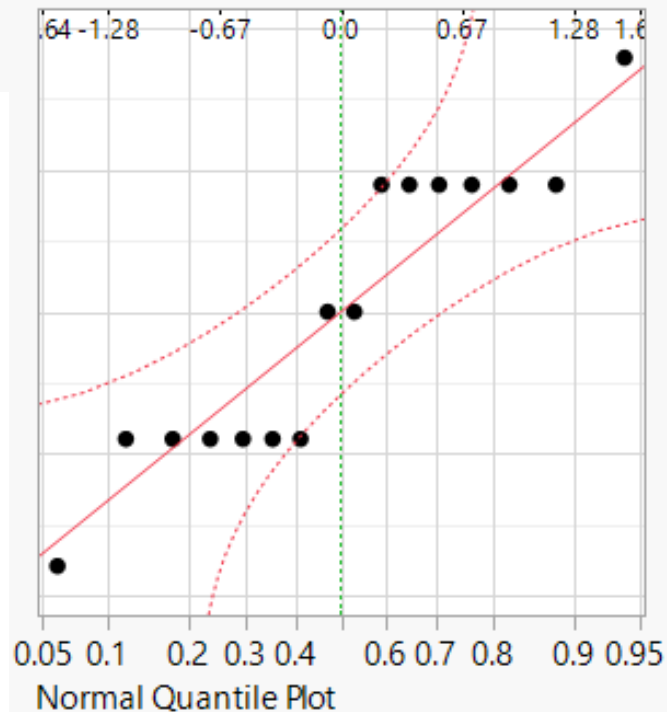
Full Model



Goodness-of-Fit Test

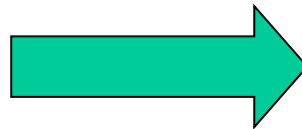
	W	Prob<W
Shapiro-Wilk	0.8743574	0.0317*
	A ²	Simulated p-Value
Anderson-Darling	1.0628016	0.0080*

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .



Don't worry

Use the **Right Model**
before trying any
transformation



Refining the Model (if necessary): Removal of nonsignificant model components

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	71.250000	23.7500	27.1429
Error	12	10.500000	0.8750	Prob > F
C. Total	15	81.750000		<.0001*

Summary of Fit

RSquare	0.87156
RSquare Adj	0.83945
Root Mean Square Error	0.935414
Mean of Response	0.875
Observations (or Sum Wgts)	16

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
CO2	1	1	30.250000	34.5714	<.0001*
Pressure	1	1	25.000000	28.5714	0.0002*
Speed	1	1	16.000000	18.2857	0.0011*

Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	4	5.500000	1.37500	2.2000
Pure Error	8	5.000000	0.62500	Prob > F
Total Error	12	10.500000		0.1592
				Max RSq
				0.9388

$$\text{Pure Error} \equiv SS_{PE} = \sum_{l=1,2^k} (n_l - 1) s_l^2 = (n - 1) \sum_{l=1,2^k} s_l^2$$

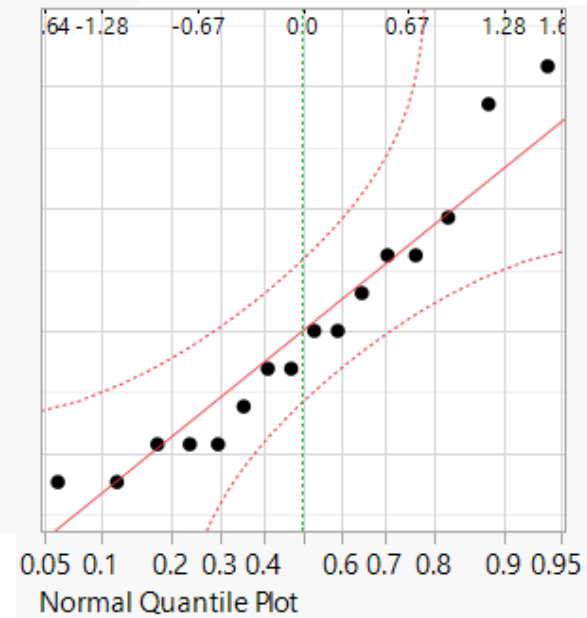
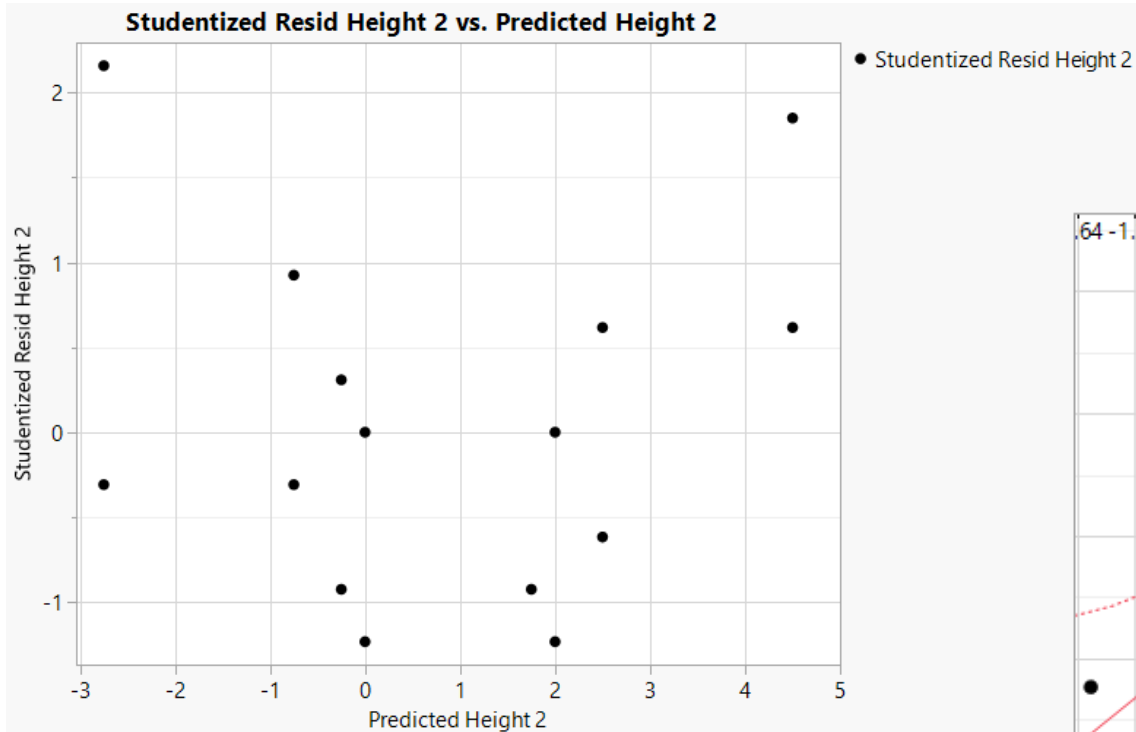
$$df_{PE} = \sum_{l=1,2^k} (n_l - 1) = 2^k (n - 1)$$

$$SS_E = SS_{PE} + SS_{LOF}$$

$$\text{Lack of fit} \equiv SS_{LOF} = SS_E - SS_{PE}$$

$$df_{LOF} = df_E - df_{PE}$$

Residual Analysis (Reduced Model)

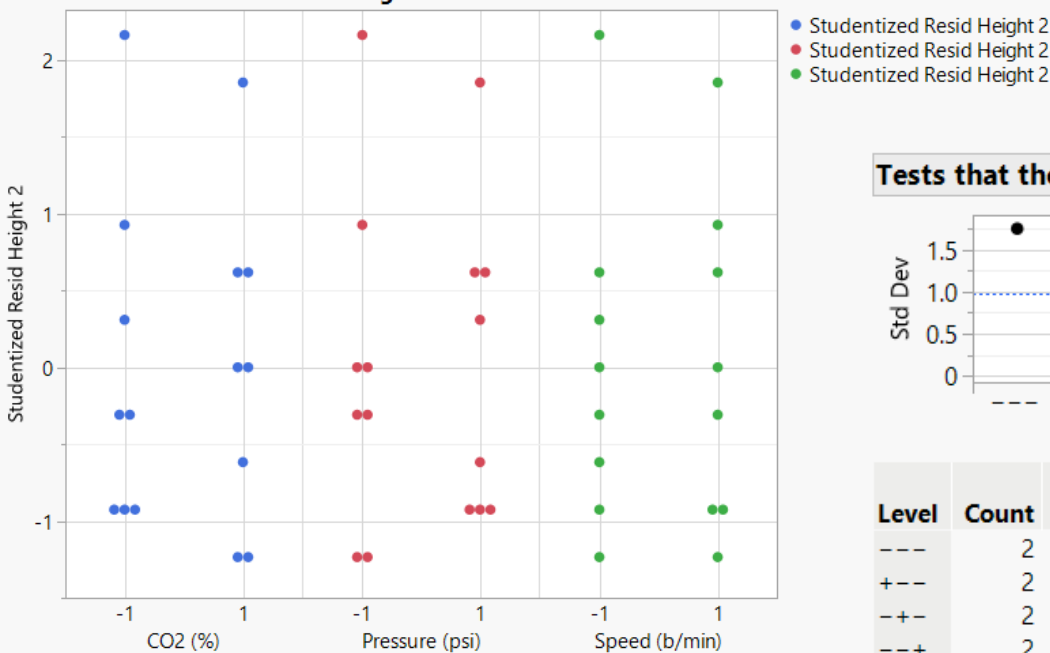


Goodness-of-Fit Test

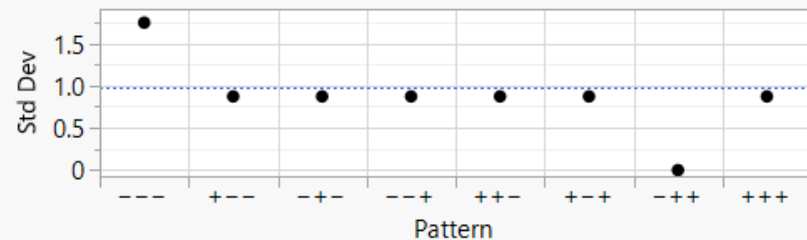
	W	Prob < W
Shapiro-Wilk	0.9210554	0.1754
	A ²	Simulated p-Value
Anderson-Darling	0.4188885	0.3052

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .

Studentized Resid Height 2 vs. CO2 & 2 more



Tests that the Variances are Equal



Level	Count	Std Dev	MeanAbsDif to Mean	MeanAbsDif to Median
---	2	1.745743	1.234427	1.234427
+--	2	0.872872	0.617213	0.617213
--+	2	0.872872	0.617213	0.617213
---	2	0.872872	0.617213	0.617213
++-	2	0.872872	0.617213	0.617213
---+	2	0.872872	0.617213	0.617213
---+	2	0.872872	0.617213	0.617213
---	2	0.000000	0.000000	0.000000
---	2	0.872872	0.617213	0.617213

Test	F Ratio	DFNum	DFDen	Prob > F
O'Brien[.5]	0.0000	-1	0	<.0001*
Brown-Forsythe	2.614e+15	7	8	<.0001*
Levene	2.614e+15	7	8	<.0001*
Bartlett	.	7	.	.

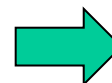
Warning: Small sample sizes. Use Caution.

Welch's Test

Welch Anova testing Means Equal, allowing Std Devs
Not Equal

F Ratio	DFNum	DFDen	Prob > F
0.5657	6	3.1037	0.7469

Use caution!
Small Sample Size n=2



Remarks (only for orthogonal designs)

If we remove some model components

$$EL \equiv \{ \text{Removed Model Components} \}$$

$$\text{i.e. } EL \equiv \{ B, AB, ABC, BC \}$$

The ANOVA Table changes only on the **Error Row**

$$SS_{E,New} = SS_{E,Old} + \sum_{l \in EL} SS_l$$

$$df_{E,New} = df_{E,Old} + |EL|$$

The coefficients of the reduced regression equation don't change

Spacing of levels in 2^k Designs (for quantitative Factors)

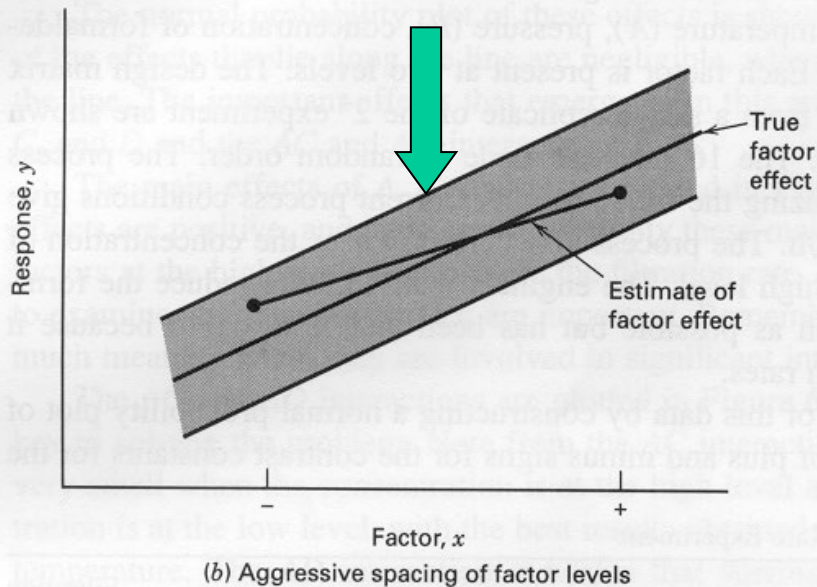
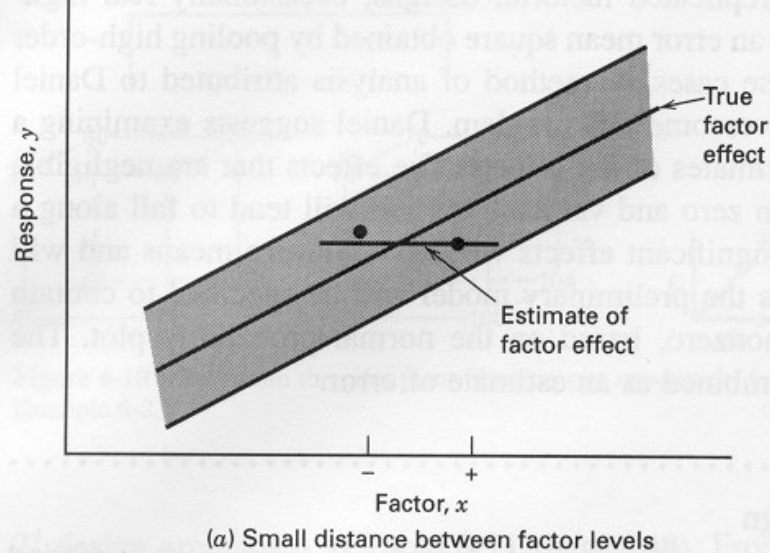


Figure 6-9 The impact of the choice of factor levels in an unreplicated design.

If factors have a small distance it's probable that the useful signal is masked by the process variability.

The problem could be tackled increasing the number of replicates

When we have just one replicate the better choice is to use a more aggressive spacing

The 2^k Factorial Design

$$\binom{k}{1} = k \quad \rightarrow \text{Main Effects} \quad A, B, C, \dots, K$$

We have

$$\binom{k}{2} = \frac{k!}{2!(k-2)!} \quad \rightarrow \quad 2fi \quad AB, AC, \dots, AK, \dots, JK$$

$$\binom{k}{3} = \frac{k!}{3!(k-3)!} \quad \rightarrow \quad 3fi \quad ABC, ABD, ABK, \dots, IJK$$

$$\binom{k}{k} = 1 \quad \rightarrow \quad kfi \quad ABCD \dots K$$

$$\text{Model terms} = 2^k = (2^k - 1) + \text{general mean}$$

i.e. for $k=6 \rightarrow$ The model terms are $64+1=65$

Probably many terms are not significant and could be used to estimate the process noise

2^k Factorial Designs without Replicates

- We have only **one observation** at each hypercube vertex
- A 2^k Factorial Designs without replicates is named “**single replicate**” of a 2^k Design or **Unreplicated Factorial**
- The lack of replicates causes some **problems** in the statistical tests:
 - The replicates are used to estimate the “**pure error**” (**internal estimate** of the **Error variance**)
 - if we use the **Full Model** we don't have any *df* for Error variance estimation because we don't have any replicate
- To **manage** the problem we could:
 - **Combine** interactions to estimate the Error variance and use a **Reduced Model: How: could you hazard a guess?**
 - **Use** the **Effect Normal Probability Plot (Daniel)**
 - ➡ – **Use** different methods: **Lenth, (Bisgaard)**

An example: 2^k Design without replicates

- A 2^4 Design is used to discover the (possible) effect of four Factors on the **Filtration rate (gal/h)** of a chemical product (a resin)
- The **Factors** are A = Temperature, B = Pressure, C = Formaldehyde Concentration, D = Stirring rate
- The runs were conducted on a pilot plant

Observations

Table 6-10 Pilot Plant Filtration Rate Experiment

StdOrder	Factor				Run Label	Filtration Rate (gal/h)
	A	B	C	D		
1	—	—	—	—	(1)	45
2	+	—	—	—	<i>a</i>	71
3	—	+	—	—	<i>b</i>	48
4	+	+	—	—	<i>ab</i>	65
5	—	—	+	—	<i>c</i>	68
6	+	—	+	—	<i>ac</i>	60
7	—	+	+	—	<i>bc</i>	80
8	+	+	+	—	<i>abc</i>	65
9	—	—	—	+	<i>d</i>	43
10	+	—	—	+	<i>ad</i>	100
11	—	+	—	+	<i>bd</i>	45
12	+	+	—	+	<i>abd</i>	104
13	—	—	+	+	<i>cd</i>	75
14	+	—	+	+	<i>acd</i>	86
15	—	+	+	+	<i>bcd</i>	70
16	+	+	+	+	<i>abcd</i>	96

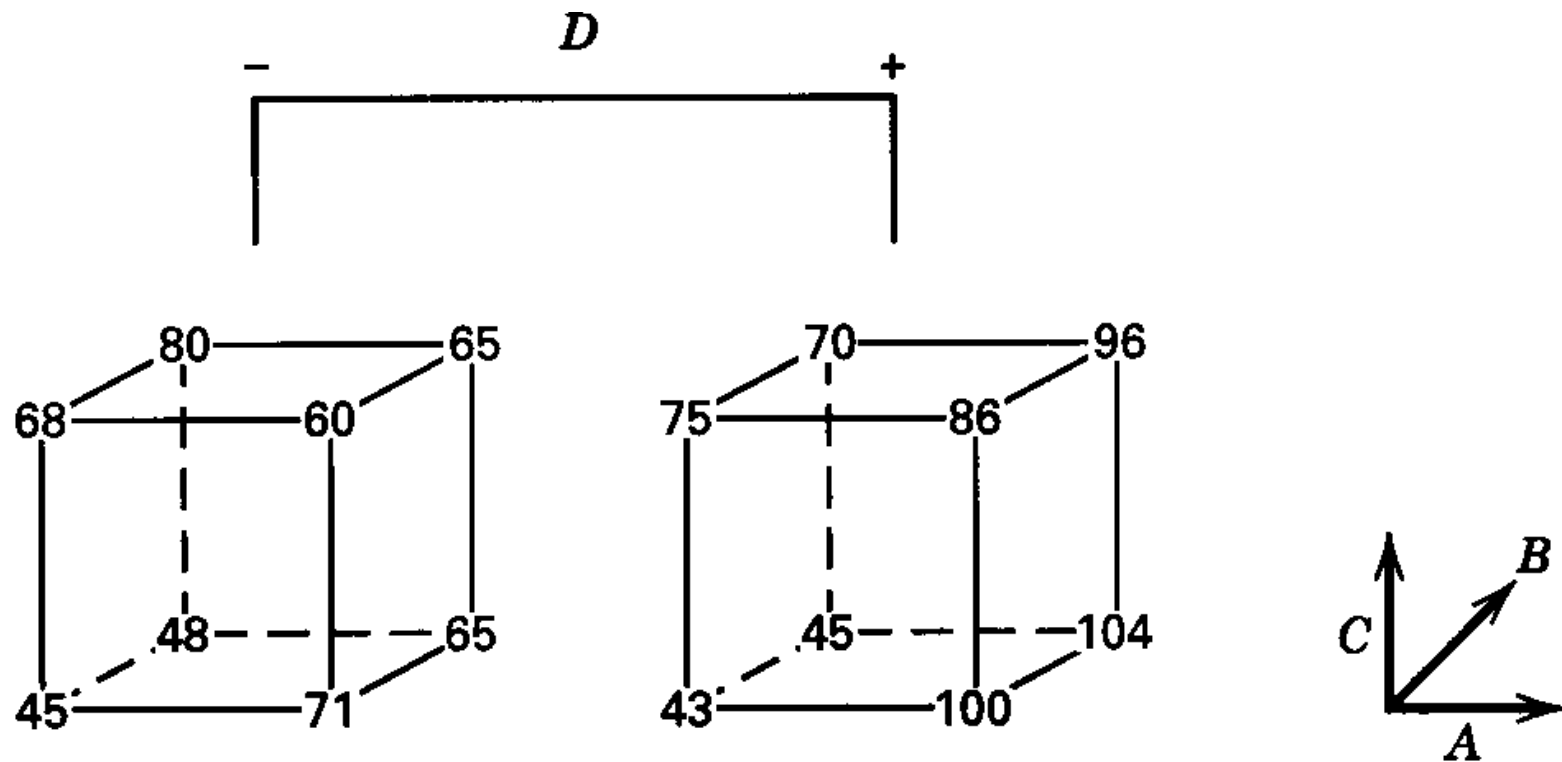
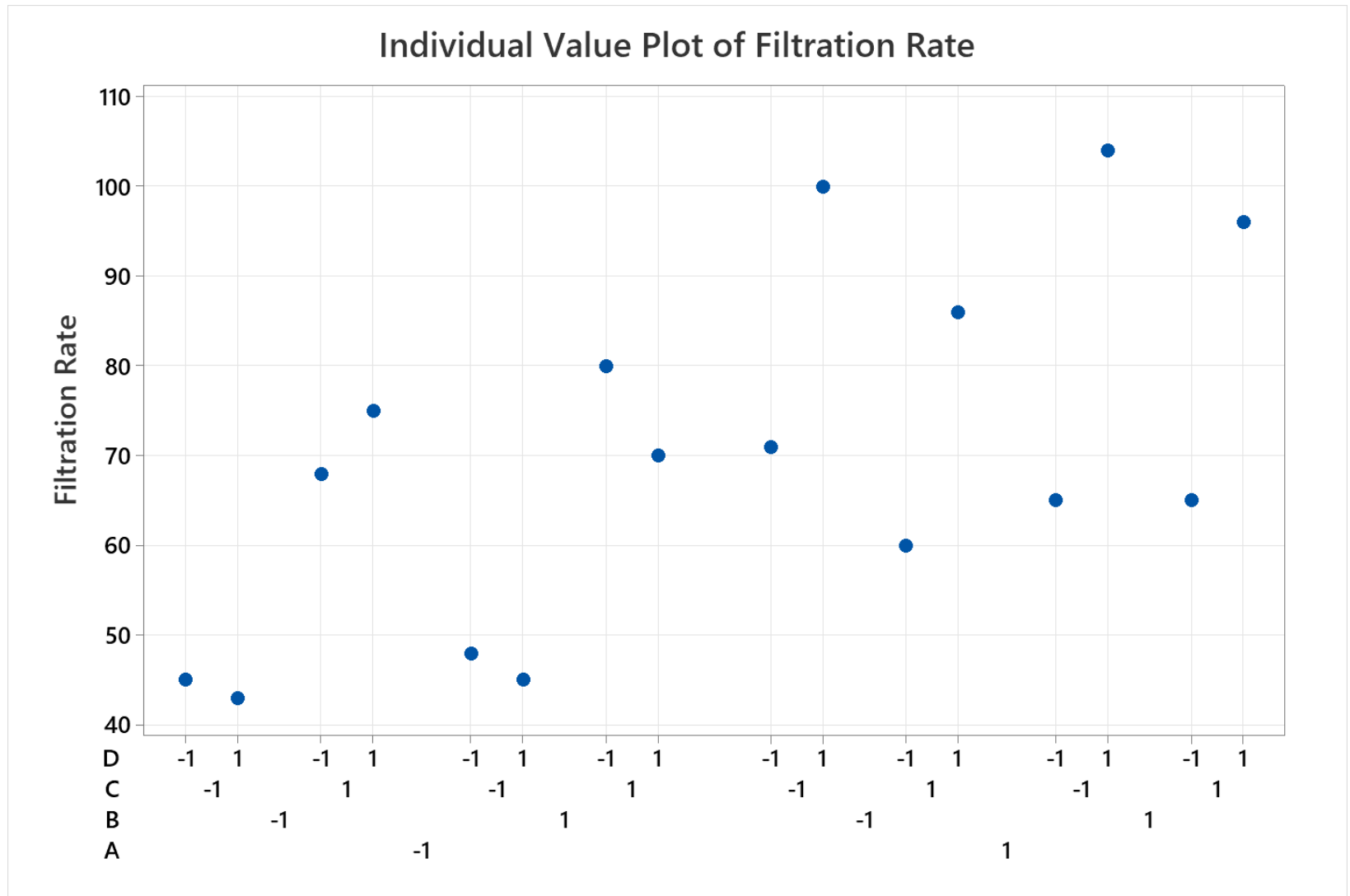
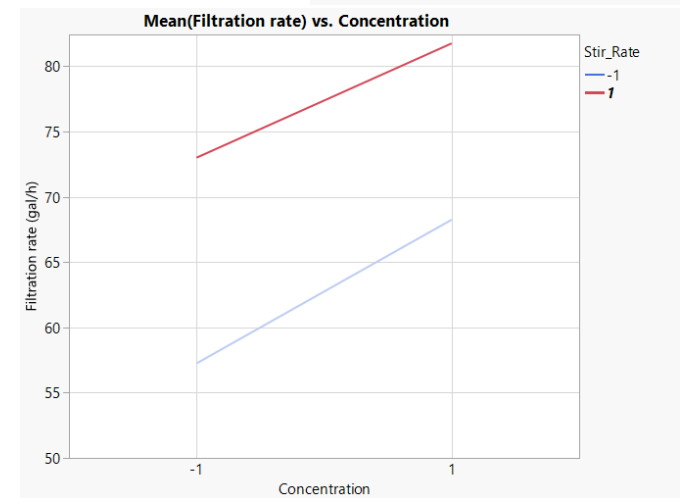
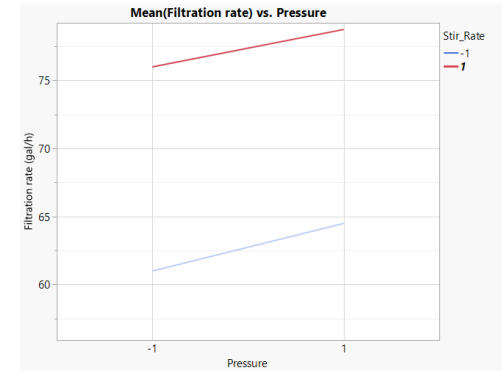
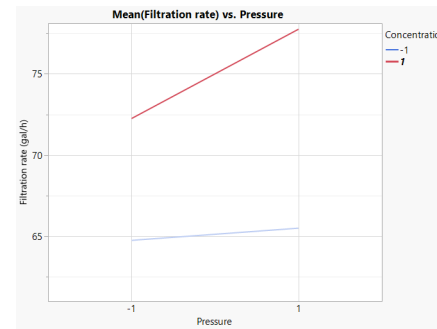
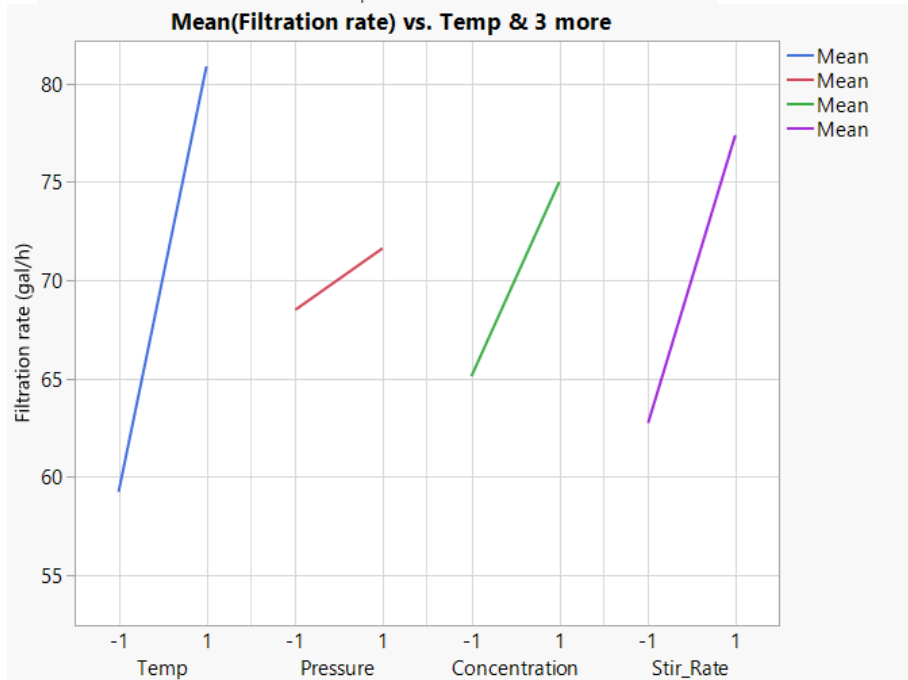
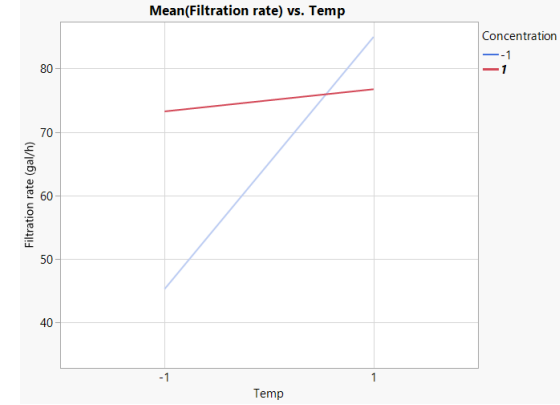
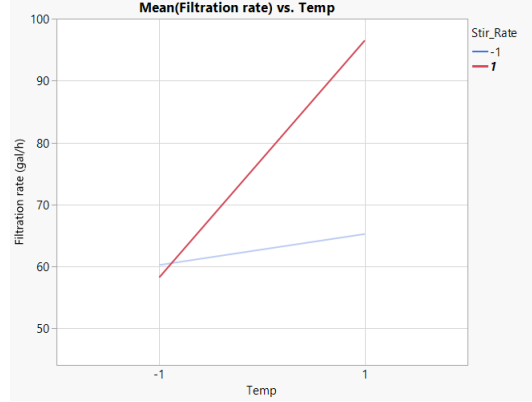
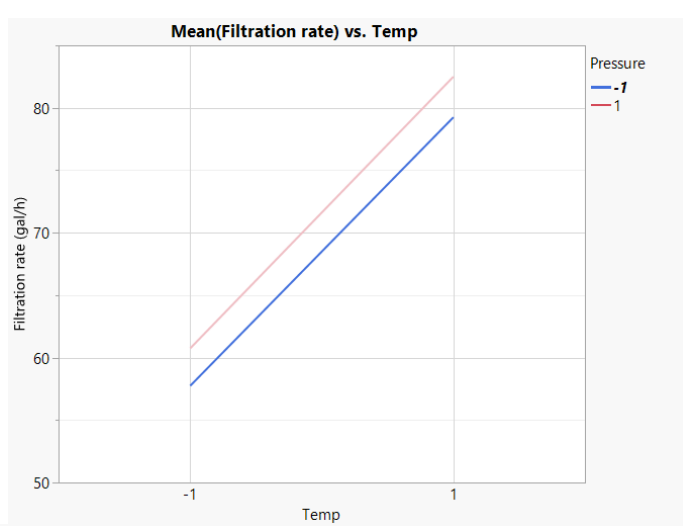


Figure 6-10 Data from the pilot plant filtration rate experiment for Example 6-2.

Individual value plot **does not make sense!**





Initial plots

10:26

QS-AA25-26

41

Effect estimates

Term	Effect	SS	% Contribution	
A	21.625	1870.56	32.64	←
B	3.125	39.0625	0.68	
C	9.875	390.062	6.81	←
D	14.625	855.563	14.93	←
AB	0.125	0.0625	0.001	
AC	-18.125	1314.06	22.93	←
AD	16.625	1105.56	19.29	←
BC	2.375	22.5625	0.39	
BD	-0.375	0.5625	0.01	
CD	-1.125	5.0625	0.089	
ABC	1.875	14.0625	0.245	
ABD	4.125	68.0625	1.188	
ACD	-1.625	10.5625	0.184	
BCD	-2.625	27.5625	0.481	
ABCD	1.375	7.5625	0.132	
TOT		5731.0		

$$\widehat{EFF} = \frac{\widehat{CONTRAST}}{n2^{k-1}}$$

$$SS = n2^{k-2} \widehat{EFF}^2$$

(in our case $k = 4$ $n = 1$)

rough approach

Include in the model **A**, **D**, **AC**, **AD** and **C**

Combine the other terms to estimate the Error variance

A = Temperature

B = Pressure

C = Formaldehyde Conc

D = Stirring rate

Effect Normal Probability Plot (Daniel)

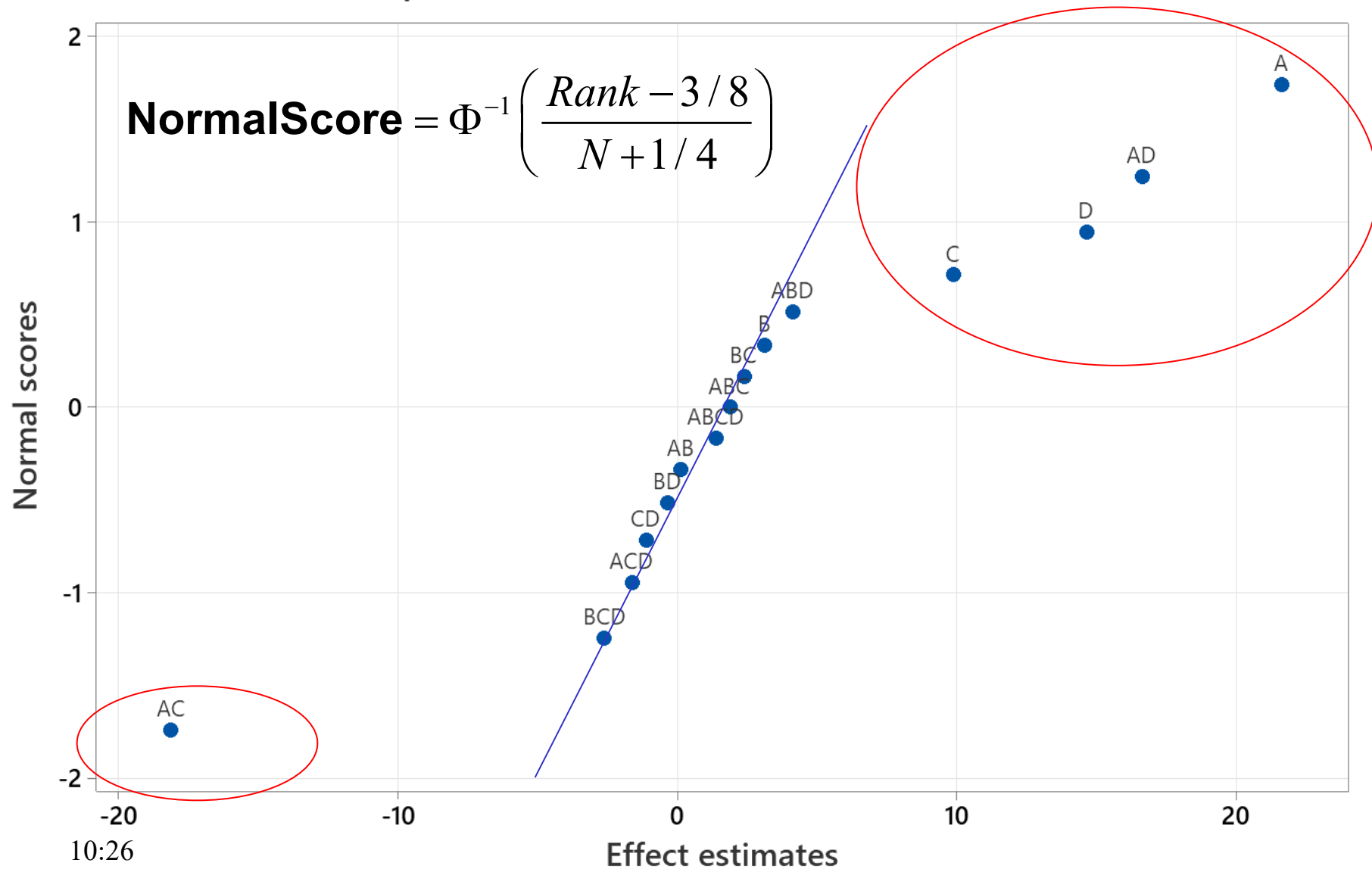
- Effect Distribution
$$\widehat{EFF}_{[]} = \frac{\widehat{CONTRAST}_{[]}}{n2^{k-1}}$$

If H_0 is true $\rightarrow y = \mu + \varepsilon$

$$\left\{ \begin{array}{l} \widehat{EFF}_{[]} \sim \text{Normal} \\ E\left(\widehat{EFF}_{[]}\right) = 0 \\ V\left(\widehat{EFF}_{[]}\right) = \frac{1}{n2^{k-2}} \sigma^2 \end{array} \right. \Rightarrow \widehat{EFF}_{[]} \sim N\left(0, \frac{1}{n2^{k-2}} \sigma^2\right)$$

Effect Normal Probability Plot(Daniel)

Scatterplot of Normal scores vs Effect estimates



Lenth's Method

- **Empirical /numerical** method to produce a **tentative list** of significative effects
- Use an estimate of the **Error variance** by combining small effects
- Some corrections have been proposed
- Useful together with the Normal Probability Plot of the Effects

Lenth Russel V. «Quick and easy analysis of Unreplicated Factorials» Technometrics November 1989 Vol 31, No.4

- Let us consider a **2^k Design without replicates**
- We are interested in finding the significative terms (we have **$m=2^k-1$ terms**, we don't consider the general mean)
- Let's compute the **m effects**

$$s_0 = 1.5 \text{ Median} \left(\left| \widehat{EFF}_j \right| \right)$$

$$PSE = 1.5 \text{ Median} \left(\left| \widehat{EFF}_j \right| ; \left| \widehat{EFF}_j \right| < 2.5s_0 \right)$$

Pseudo Standard Error Lenth proves that PSE is a “reasonable” estimator of the **Standard deviation** of an **Effect** when we have few significative terms

- The method uses two criteria (the second criteria considers the familywise error)

$$ME = t_{\alpha/2}(d)PSE \quad d = \frac{m}{3}$$

Margin of Error

$$SME = t_{\gamma}(d)PSE \quad \gamma = \left[1 - (1 - \alpha_{FAM})^{1/m} \right] / 2 \simeq \frac{\alpha_{FAM}}{2m}$$

Simultaneous Margin of Error

Correction of constants

- **Ye e Hamada** (2000) proposed to correct the Lenth's constants with the ones in the following table ($\alpha=5\%$)

Number of Effects	7	15	31
Original for ME	3.764	2.571	2.218
Corrected for ME	2.30	2.16	2.06
Original for SME	9.008	5.219	4.218
Corrected for SME	4.87	4.24	3.92

Example of the Lenth's Method

		Estimate	Effect
1	AB	0.125	0.125
2	BD	-0.375	0.375
3	CD	-1.125	1.125
4	ABCD	1.375	1.375
5	ACD	-1.625	1.625
6	ABC	1.875	1.875
7	BC	2.375	2.375
8	BCD	-2.625	2.625
9	B	3.125	3.125
10	ABD	4.125	4.125
11	C	9.875	9.875
12	D	14.625	14.625
13	AD	16.625	16.625
14	AC	-18.125	18.125
15	A	21.625	21.625

$$s_0 = 1.5 * |-2.625| = 3.9375$$

$$2.5s_0 = 9.844$$

$$\frac{1.625 + 1.875}{2} = 1.75$$

$$PSE = 1.5 * |1.75| = 2.625$$

$$\gamma = (1 - 0.95^{1/15}) / 2 = 0.001707$$

$$t_{2,5\%}(5) = 2.571 \quad t_{\gamma\%}(5) = 5.219$$

$$ME = t_{2,5\%}(5) * 2.625 = 2.16 * 2.625 = 5.67$$

$$SME = t_{\gamma\%}(5) * 2.625 = 4.24 * 2.625 = 11.13$$

If we do not use the Y-H corrections

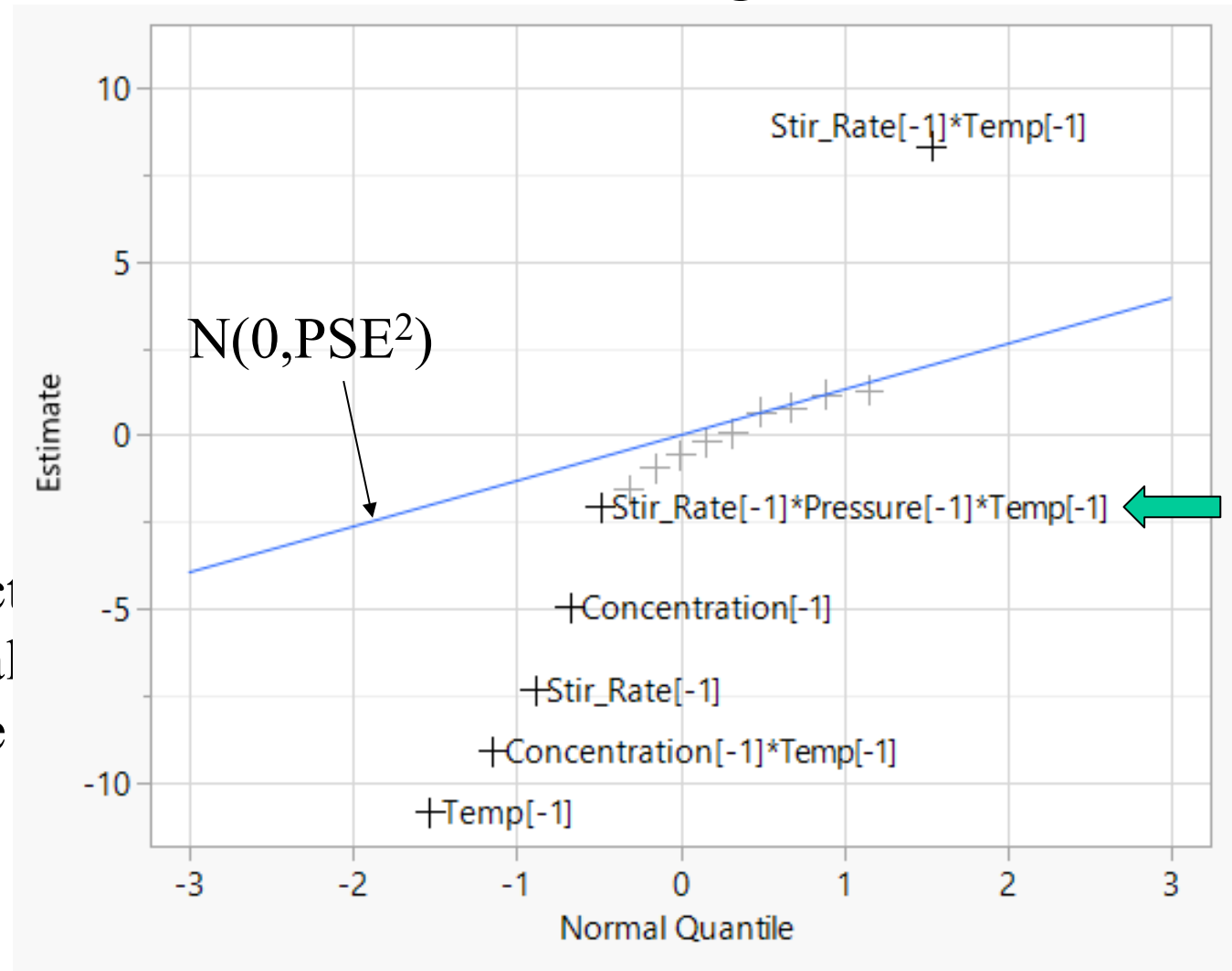
$$ME = 2.571 * 2.625 = 6.75$$

$$SME = 5.219 * 2.625 = 13.70$$

Using the correction

Effect Normal Probability Plot(Daniel)

Command **JMP->Fit Model >Effect Screening**



JMP gives half effect
and consequently hal
PSE (in the example
PSE= 1.3125)

ANOVA Table – Reduced Model

Summary of Fit

RSquare	0.965952
RSquare Adj	0.948929
Root Mean Square Error	4.417296
Mean of Response	70.0625
Observations (or Sum Wgts)	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	5535.8125	1107.16	56.7412
Error	10	195.1250	19.51	Prob > F
C. Total	15	5730.9375		<.0001*

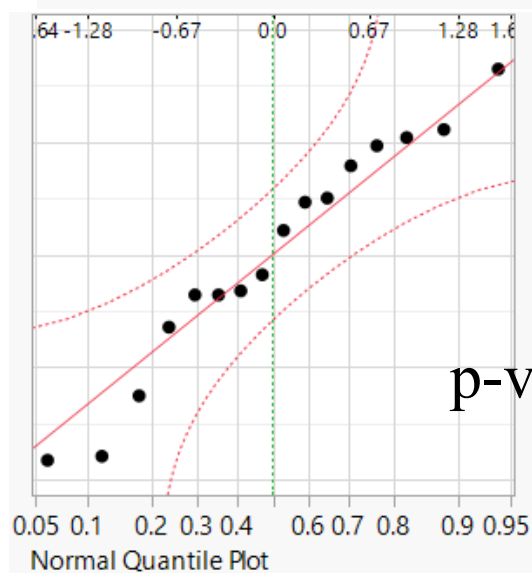
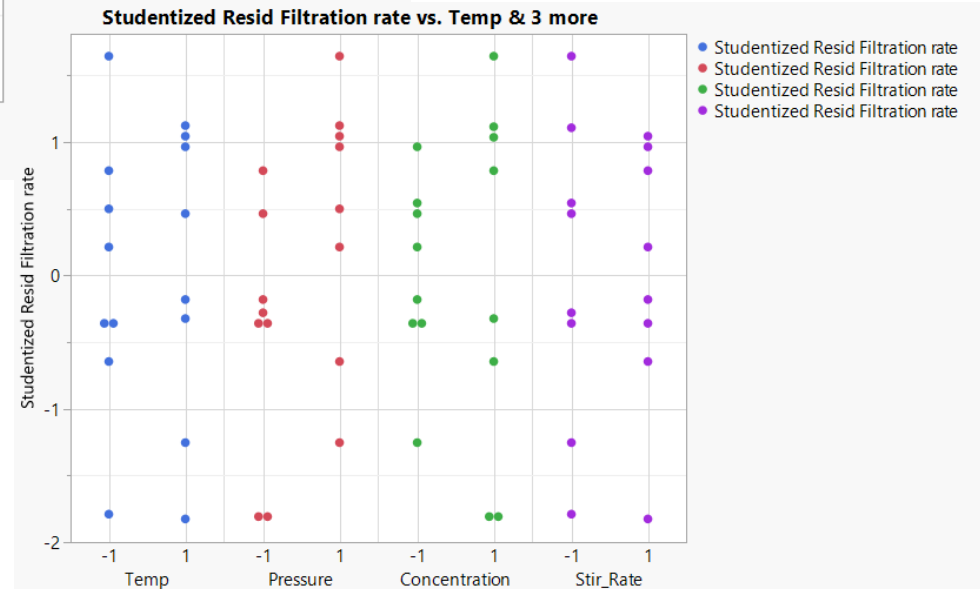
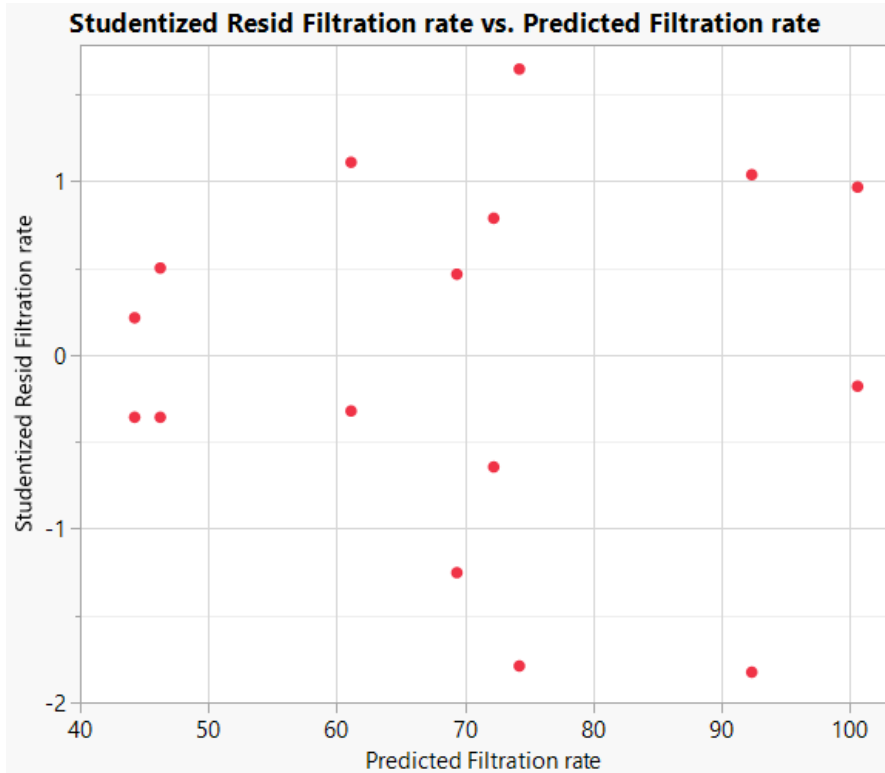
Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	2	15.62500	7.8125	0.3482
Pure Error	8	179.50000	22.4375	Prob > F
Total Error	10	195.12500		0.7162
				Max RSq
				0.9687

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Stir_Rate	1	1	855.5625	43.8469	<.0001*
Concentration	1	1	390.0625	19.9904	0.0012*
Temp	1	1	1870.5625	95.8648	<.0001*
Stir_Rate*Temp	1	1	1105.5625	56.6592	<.0001*
Concentration*Temp	1	1	1314.0625	67.3447	<.0001*

Residual analysis



If **one** Factor is removed, the 2^4 design without replicates **projects** in a 2^3 Design with **two replicates**

For this reason we can make a test on variances

The projection property is very useful and suggests to use **Fractional Factorial Plans**

Regression Model

Parameter Estimates					
Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	70.0625	1.104324	63.44	<.0001*	.
Stir_Rate[-1]	-7.3125	1.104324	-6.62	<.0001*	1
Concentration[-1]	-4.9375	1.104324	-4.47	0.0012*	1
Temp[-1]	-10.8125	1.104324	-9.79	<.0001*	1
Stir_Rate[-1]*Temp[-1]	8.3125	1.104324	7.53	<.0001*	1
Concentration[-1]*Temp[-1]	-9.0625	1.104324	-8.21	<.0001*	1

The regression equation is:

Filtration rate= 70.1 + 10.8 Temp + 4.94 Concentration + 7.31 Stir_Rate+8.31 Stir_Rate*Temp - 9.06 Concentration*Temp

Example 6-3 (data transformation)

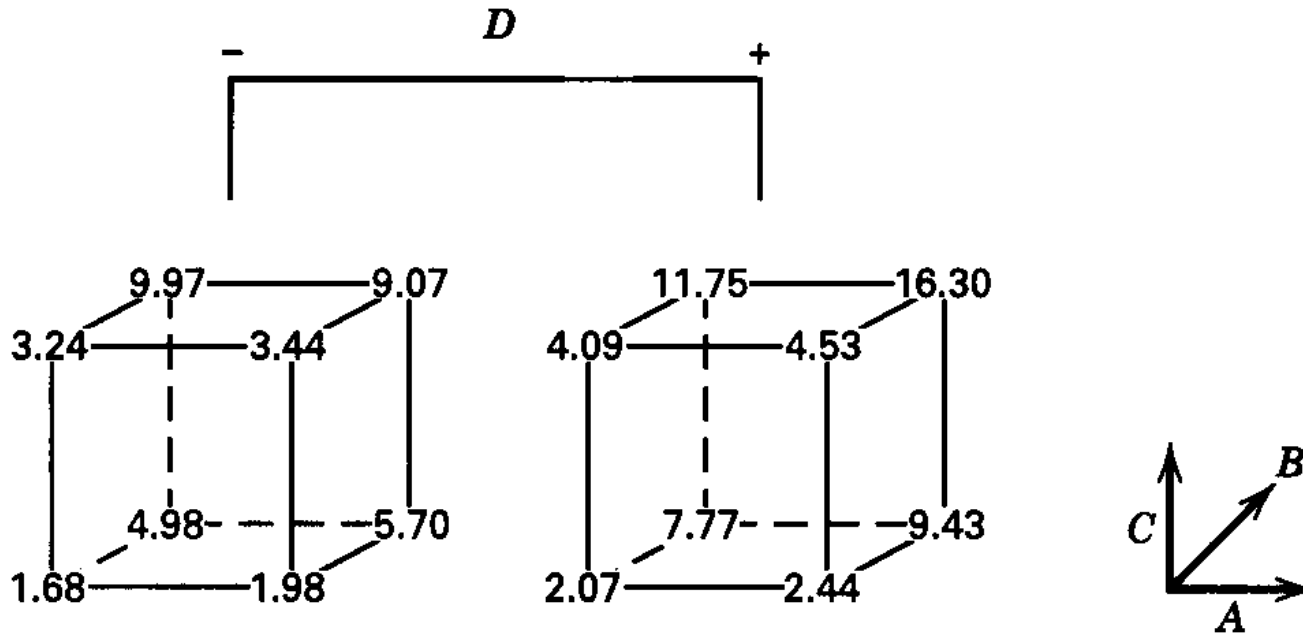
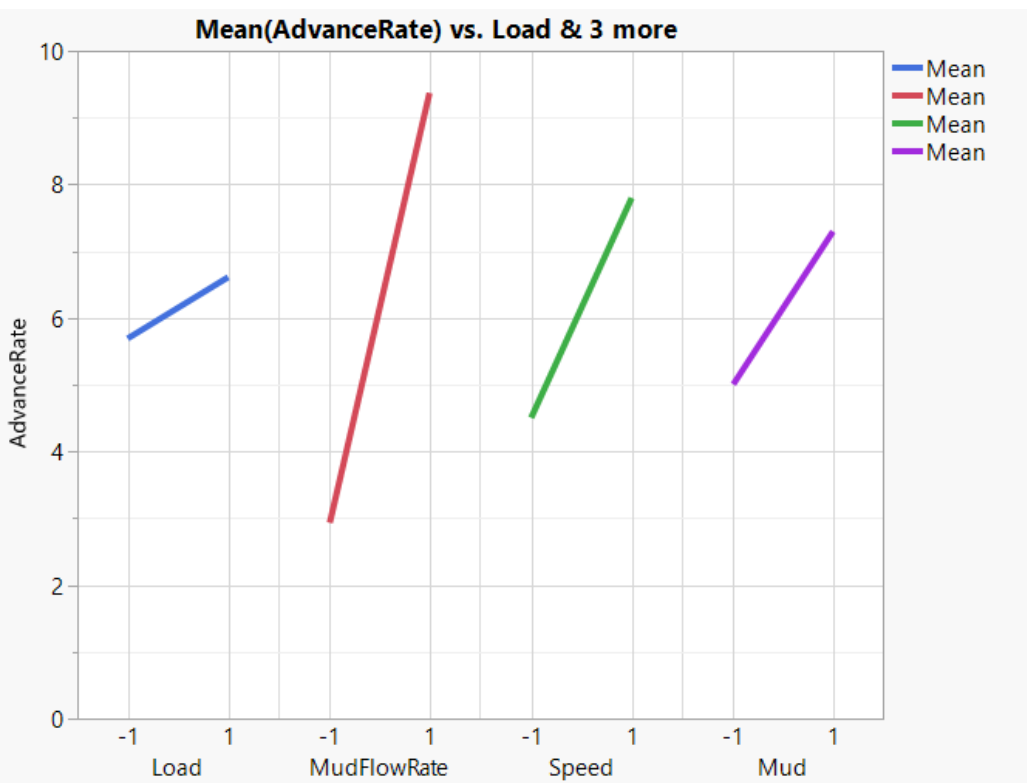


Figure 6-17 Data from the drilling experiment of Example 6-3.

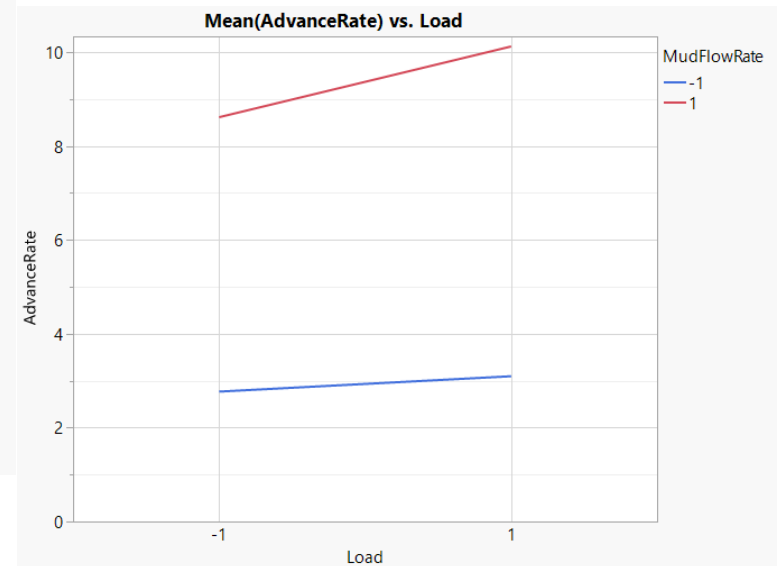
Response variable: y = Advance rate of a drill

Factors: A = Drill load, B = Mud flow rate, C = Rotational speed, D = Type of drilling mud



Usual plots

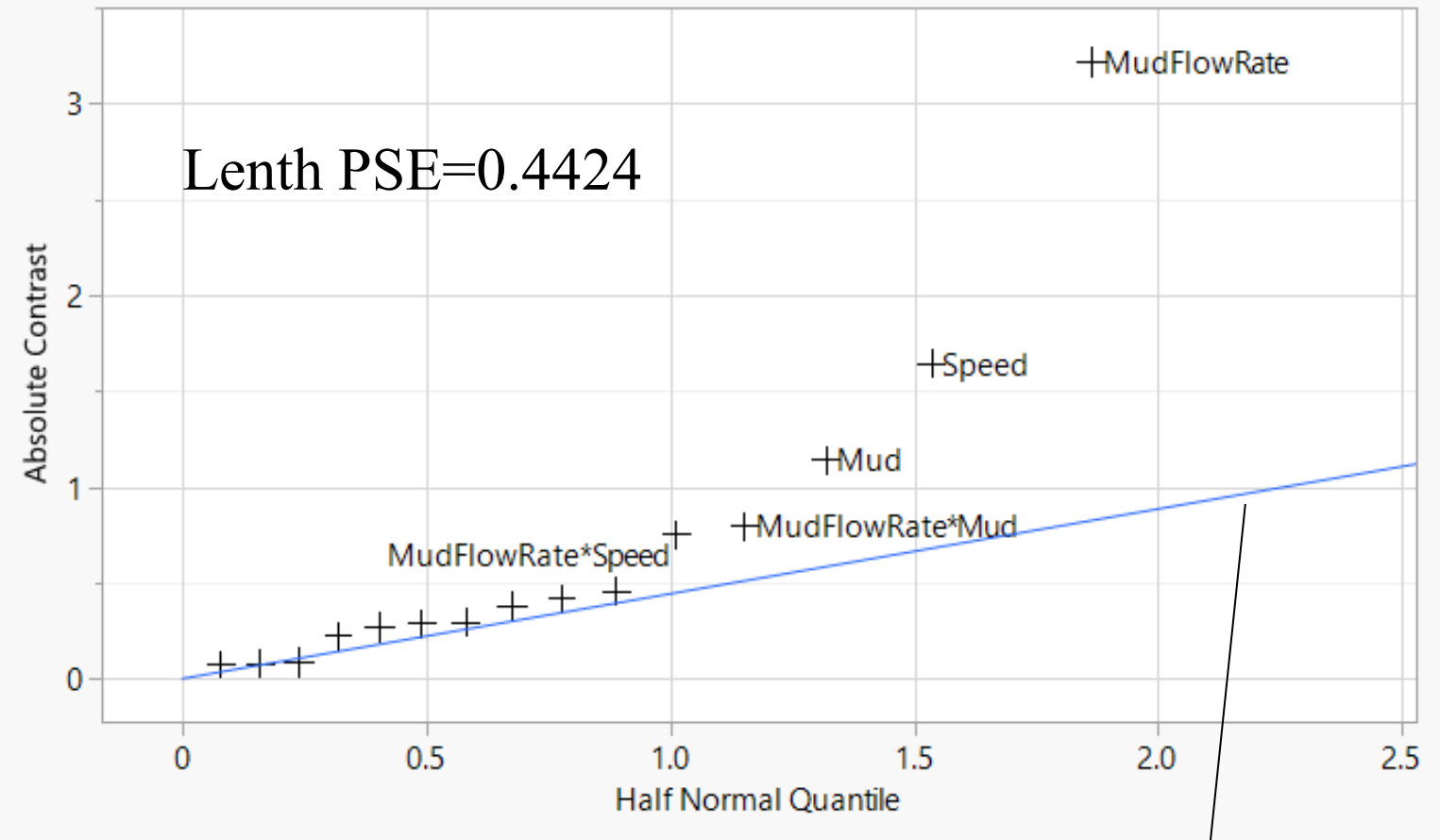
remark: no replicates



- A Drill load
- B Mud flow rate
- C Rotational speed
- D Drilling Mud

no apparent interactions in all
interaction plot

Lenth's Method + Daniel's Plot



$N(0, PSE^2)$

Analysis of Variance

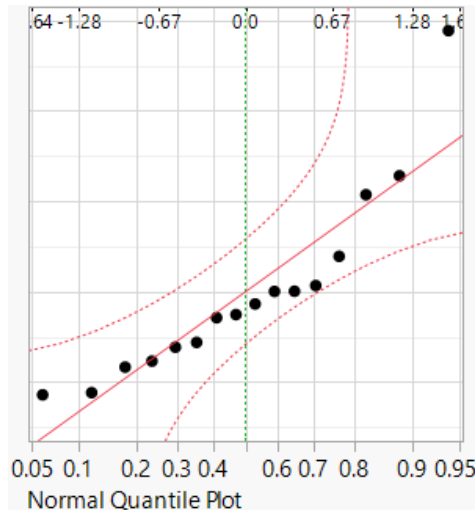
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	230.10425	76.7014	28.0515
Error	12	32.81165	2.7343	Prob > F
C. Total	15	262.91590		<.0001*

Summary of Fit

RSquare	0.875201
RSquare Adj	0.844001
Root Mean Square Error	1.653573
Mean of Response	6.1525
Observations (or Sum Wgts)	16

Effect Tests

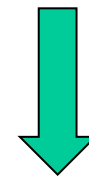
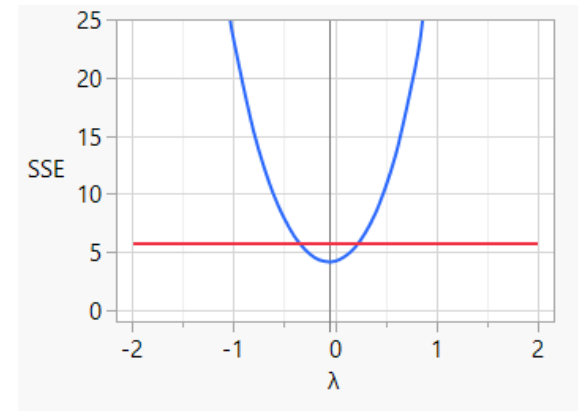
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
MudFlowRate	1	1	165.76563	60.6244	<.0001*
Speed	1	1	43.36223	15.8586	0.0018*
Mud	1	1	20.97640	7.6716	0.0170*



Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.859697	0.0190*
	A ²	Simulated p-Value
Anderson-Darling	0.7493247	0.0372*

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.



$$y^* = \text{Ln } y$$

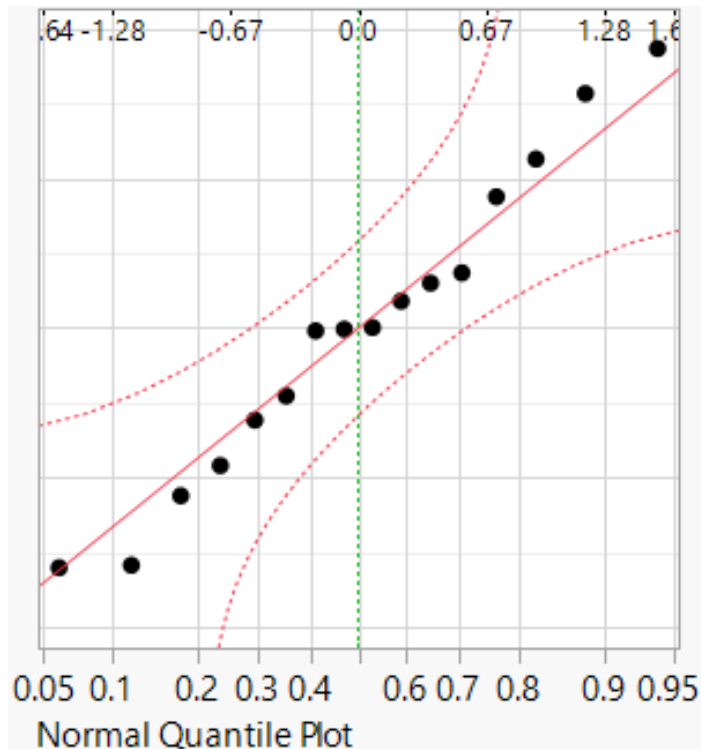
ANOVA Table for transformed data

Summary of Fit

RSquare	0.976306
RSquare Adj	0.970383
Root Mean Square Error	0.119952
Mean of Response	1.596961
Observations (or Sum Wgts)	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	7.1146186	2.37154	164.8223
Error	12	0.1726616	0.01439	Prob > F
C. Total	15	7.2872802		<.0001*



Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
MudFlowRate	1	1	5.3452105	371.4928	<.0001*
Speed	1	1	1.3388707	93.0517	<.0001*
Mud	1	1	0.4305374	29.9224	0.0001*

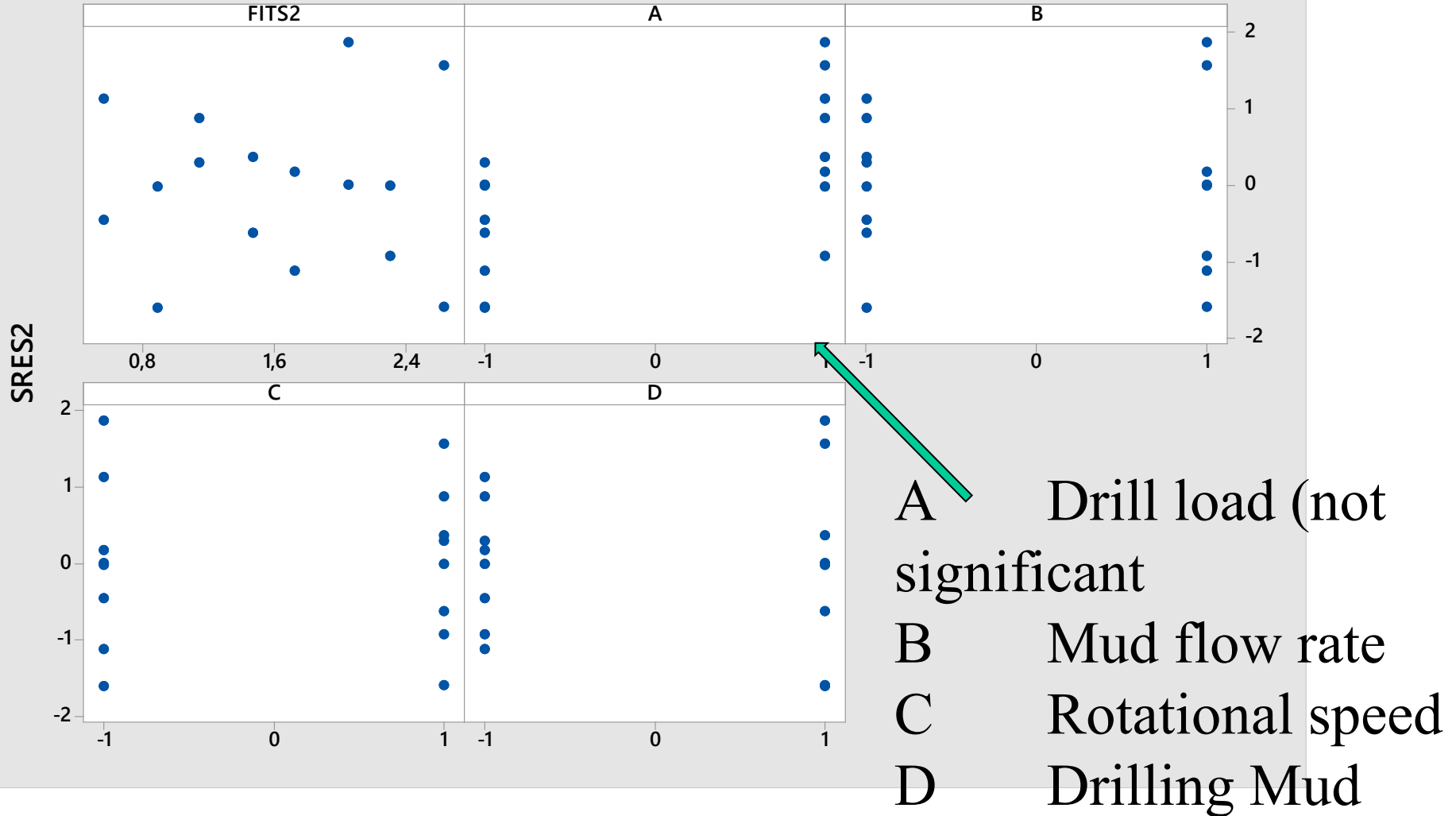
Goodness-of-Fit Test

	W	Prob < W
Shapiro-Wilk	0.9681449	0.8077
	A ²	Simulated p-Value
Anderson-Darling	0.1888081	0.8948

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

Residual analysis (transformed data)

Scatterplot of SRES2 vs FITS2; A; B; C; D

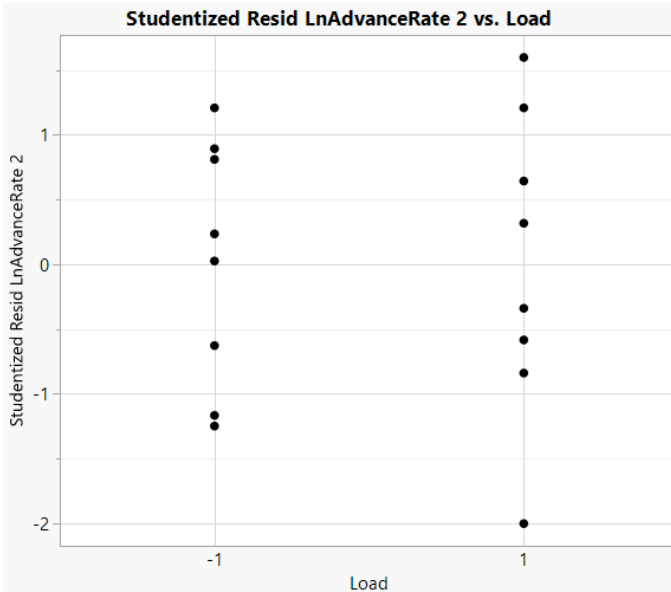


A possible refinement

Summary of Fit

RSquare	0.985802
RSquare Adj	0.980639
Root Mean Square Error	0.096985
Mean of Response	1.596961
Observations (or Sum Wgts)	16

R^2_{Adj} from 97.04 to 98.06%



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	4	7.1838130	1.79595	190.9349
Error	11	0.1034671	0.00941	Prob > F
C. Total	15	7.2872802		<.0001*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Load	1	1	0.0691944	7.3563	0.0202*
MudFlowRate	1	1	5.3452105	568.2704	<.0001*
Speed	1	1	1.3388707	142.3406	<.0001*
Mud	1	1	0.4305374	45.7721	<.0001*

Residual check OK

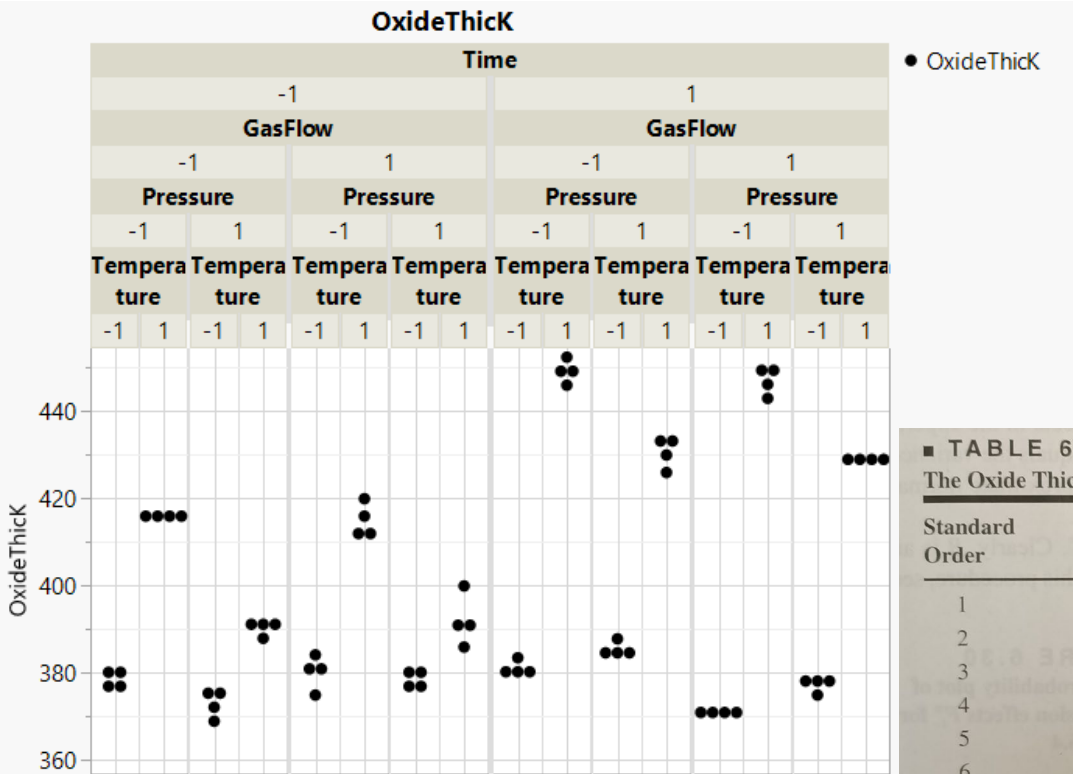
The Model using transformed data

- The “**Ln**” model is better than the previous one ?
- Generally speaking we will prefer a **Simpler Model** in a transformed scale than a **More Complicated Model** in the original scale
- Sometimes the transformation suggests the physical model laying behind the observations

Example 6.5 (Repeated Measurements)

- The response is the **oxide thickness** on a wafer.
- The factors are
 - A Temperature
 - B Time
 - C Pressure
 - D Gas flow
- 2^4 factorial experiment was run
- **In each run 4 wafers** were stacked in a special fixture in the oxidation furnace.
- At the end of the run, the **oxide thickness** is measured for each wafer.
- Discuss the experiment conduction

Data and Initial Plots

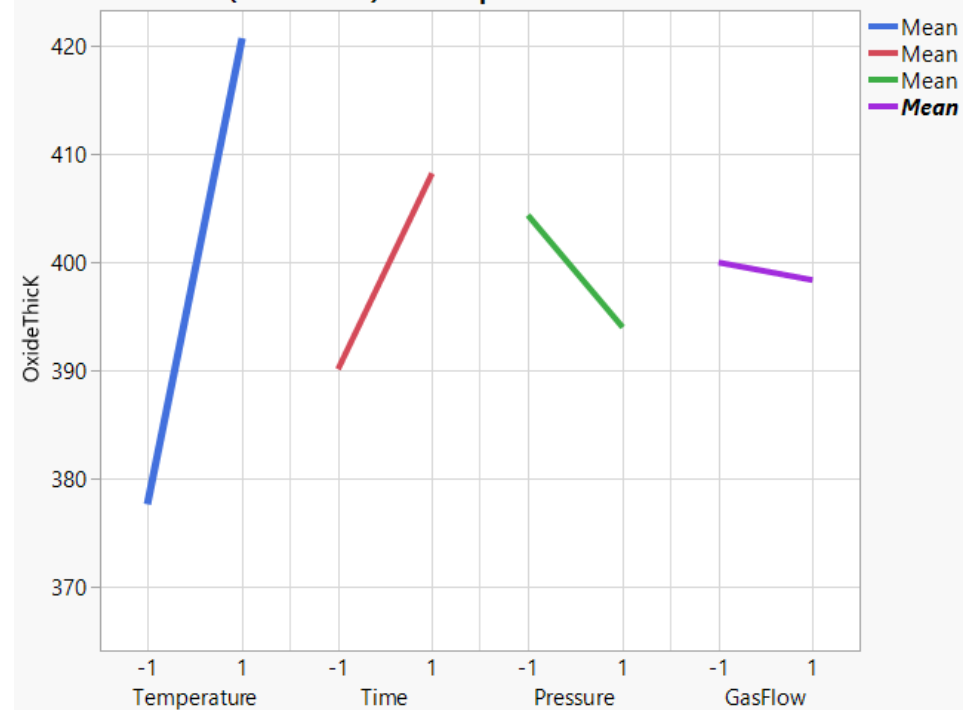


■ TABLE 6.18

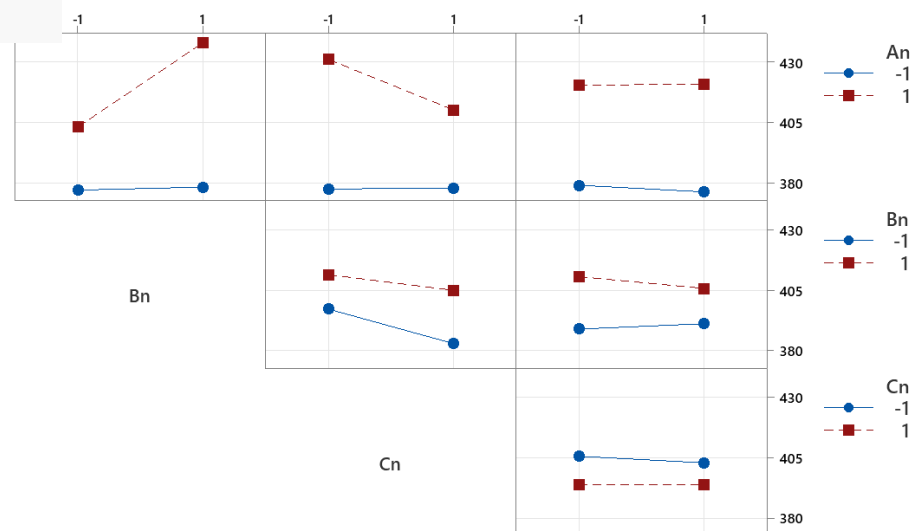
The Oxide Thickness Experiment

Standard Order	Run Order	A	B	C	D	Thickness			
1	10	-1	-1	-1	-1	378	376	379	379
2	7	1	-1	-1	-1	415	416	416	417
3	3	-1	1	-1	-1	380	379	382	383
4	9	1	1	-1	-1	450	446	449	447
5	6	-1	-1	1	-1	375	371	373	369
6	2	1	-1	1	-1	391	390	388	391
7	5	-1	1	1	-1	384	385	386	385
8	4	1	1	1	-1	426	433	430	431
9	12	-1	-1	-1	1	381	381	375	383
10	16	1	-1	-1	1	416	420	412	412
11	8	-1	1	-1	1	371	372	371	370
12	1	1	1	-1	1	445	448	443	448
13	14	-1	-1	1	1	377	377	379	379
14	15	1	-1	1	1	391	391	386	400
15	11	-1	1	1	1	375	376	376	377
16	13	1	1	1	1	430	430	428	428

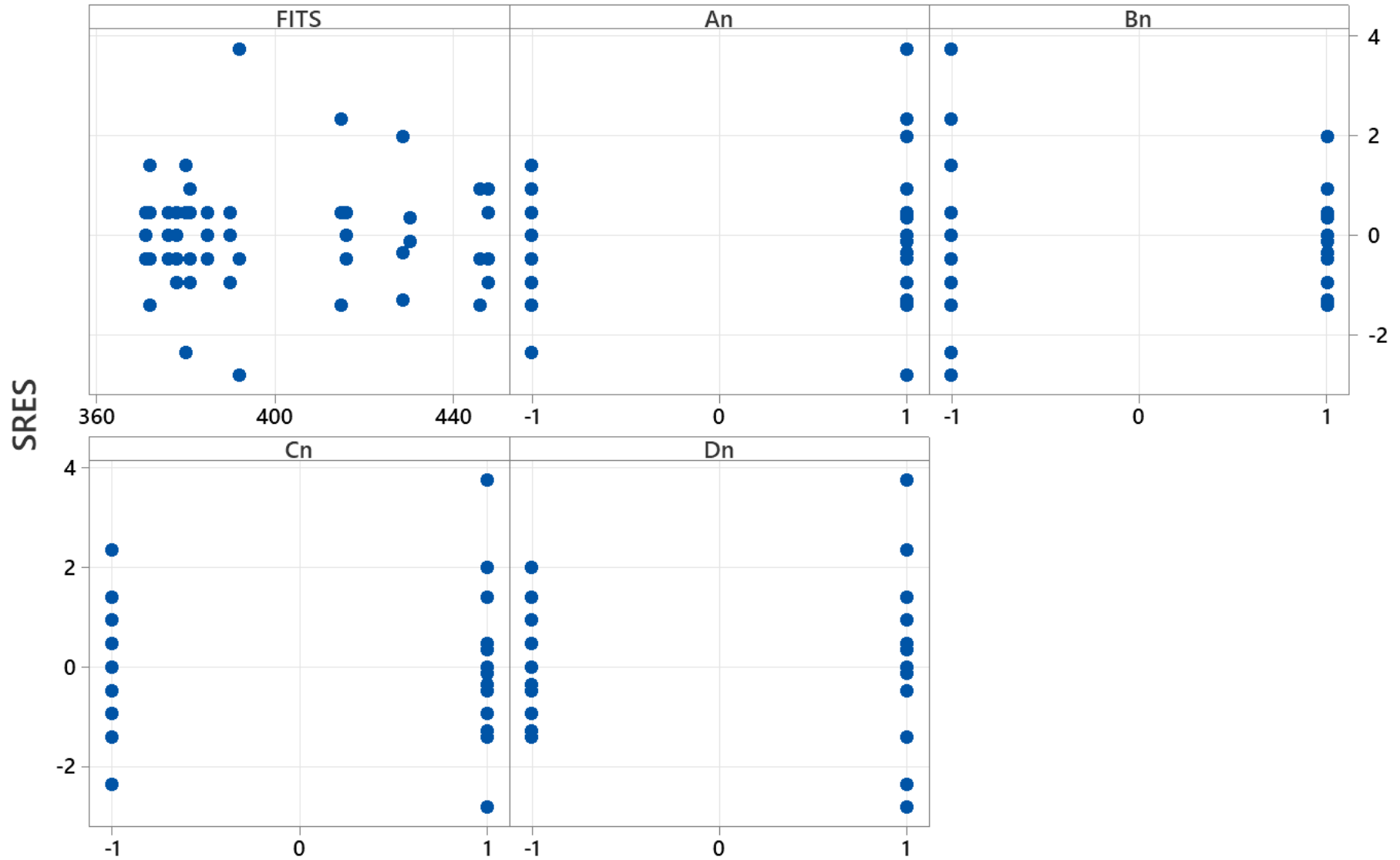
Mean(OxideThick) vs. Temperature & 3 more

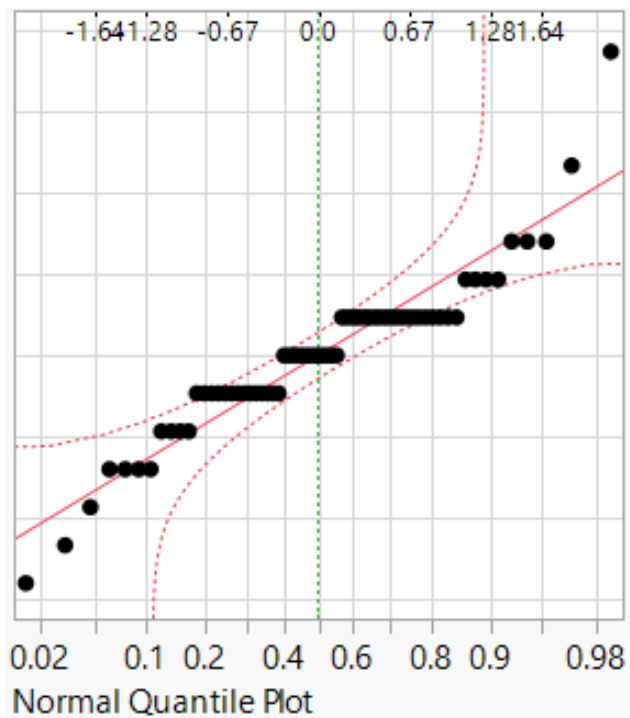


Interaction Plot for t
Data Means



Scatterplot of SRES vs FITS; An; Bn; Cn; Dn



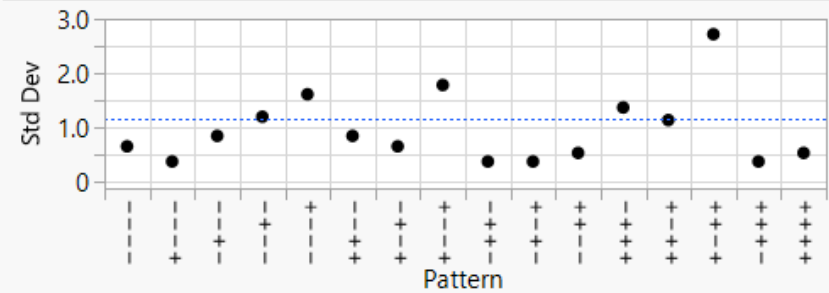


Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9263834	0.0009*
	A ²	Simulated p-Value
Anderson-Darling	1.6715515	0.0004*

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

Tests that the Variances are Equal



Level	Count	Std Dev	MeanAbsDif to Mean	MeanAbsDif to Median
----	4	0.659829	0.466569	0.466569
+---	4	0.380952	0.233285	0.233285
-+--	4	0.851835	0.699854	0.699854
--+-	4	1.204677	0.933139	0.933139
---+	4	1.616244	1.166424	0.933139
++--	4	0.851835	0.699854	0.699854
+-+-	4	0.659829	0.466569	0.466569
+-+-	4	1.786825	1.399708	1.399708
-++-	4	0.380952	0.233285	0.233285
-+-+	4	0.380952	0.233285	0.233285
--++	4	0.538748	0.466569	0.466569
+++--	4	1.373543	0.933139	0.933139
+++-	4	1.142857	0.933139	0.933139
+++	4	2.720544	1.866278	1.632993
-+++	4	0.380952	0.233285	0.233285
++++	4	0.538748	0.466569	0.466569

Test	F Ratio	DFNum	DFDen	Prob > F
O'Brien[.5]	1.3457	15	48	0.2135
Brown-Forsythe	1.2732	15	48	0.2555
Levene	2.3494	15	48	0.0129*
Bartlett	2.0938	15	.	0.0077*

Warning: Small sample sizes. Use Caution.

Welch's Test

Welch Anova testing Means Equal, allowing Std Devs Not Equal

F Ratio	DFNum	DFDen	Prob > F
0.0000	15	18.065	1.0000

ANOVA Table Full Model

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	15	43801.750	2920.12	476.7537
Error	48	294.000	6.13	Prob > F
C. Total	63	44095.750		<.0001*

Summary of Fit

RSquare	0.993333
RSquare Adj	0.991249
Root Mean Square Error	2.474874
Mean of Response	399.1875
Observations (or Sum Wgts)	64

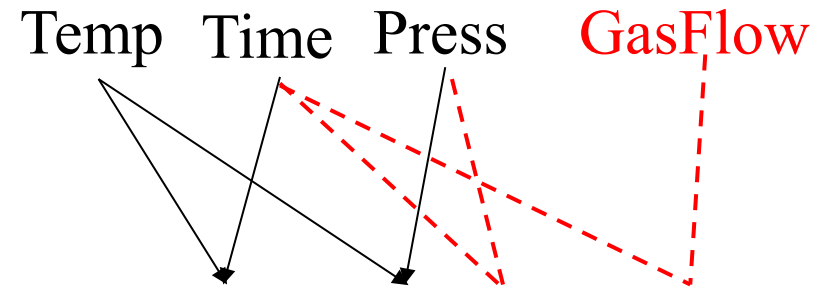
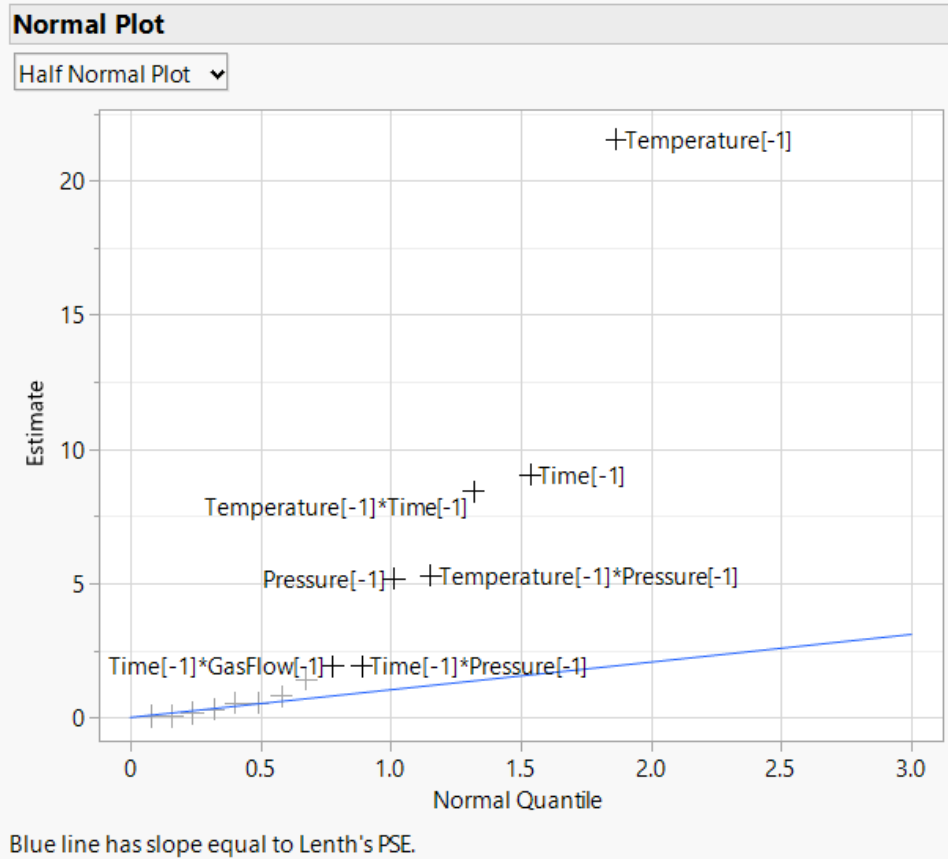
Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temperature	1	1	29756.250	4858.163	<.0001*
Time	1	1	5256.250	858.1633	<.0001*
Pressure	1	1	1722.250	281.1837	<.0001*
GasFlow	1	1	42.250	6.8980	0.0115*
Temperature*Time	1	1	4556.250	743.8776	<.0001*
Temperature*Pressure	1	1	1806.250	294.8980	<.0001*
Time*Pressure	1	1	240.250	39.2245	<.0001*
Temperature*GasFlow	1	1	20.250	3.3061	0.0753
Time*GasFlow	1	1	240.250	39.2245	<.0001*
Pressure*GasFlow	1	1	20.250	3.3061	0.0753
Temperature*Time*Pressure	1	1	2.250	0.3673	0.5473
Temperature*Time*GasFlow	1	1	132.250	21.5918	<.0001*
Temperature*Pressure*GasFlow	1	1	0.250	0.0408	0.8407
Time*Pressure*GasFlow	1	1	6.250	1.0204	0.3175
Temperature*Time*Pressure*GasFlow	1	1	0.250	0.0408	0.8407

A lot of terms seems to be significative.

Is it correct?

Average Thickness (computed on 4 wafers)



First Model:

Temp, Time, Press,
Temp*Time, Temp*Press

Second Model:

Temp, Time, Press, GasFlow
Temp*Time, Temp*Press,
Time*Press, Time*GasFlow

First Model

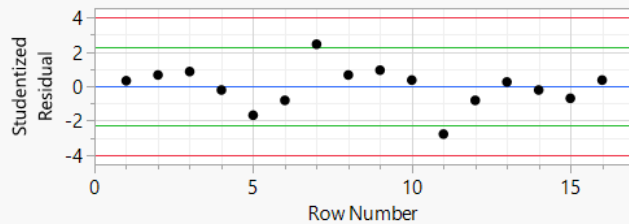
Summary of Fit

RSquare	0.983916
RSquare Adj	0.975874
Root Mean Square Error	4.196725
Mean of Response	399.1875
Observations (or Sum Wgts)	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	10774.313	2154.86	122.3485
Error	10	176.125	17.61	Prob > F
C. Total	15	10950.438		<.0001*

Studentized Residuals



Externally studentized residuals with 95% simultaneous limits (Bonferroni) in red, individual limits in green.

Effect Tests

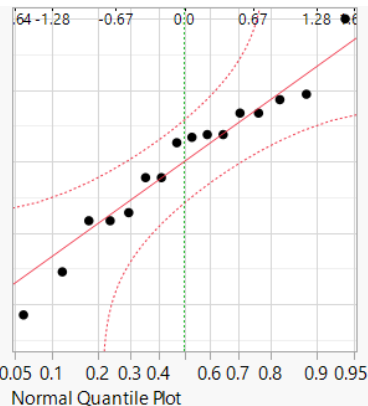
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temperature	1	1	7439.0625	422.3740	<.0001*
Time	1	1	1314.0625	74.6097	<.0001*
Pressure	1	1	430.5625	24.4464	0.0006*
Temperature*Time	1	1	1139.0625	64.6735	<.0001*
Temperature*Pressure	1	1	451.5625	25.6388	0.0005*

Goodness-of-Fit Test

	W	Prob < W
Shapiro-Wilk	0.9667212	0.7831
	A ²	Simulated p-Value
Anderson-Darling	0.3234104	0.5148

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

Other Residual
Checks of are OK

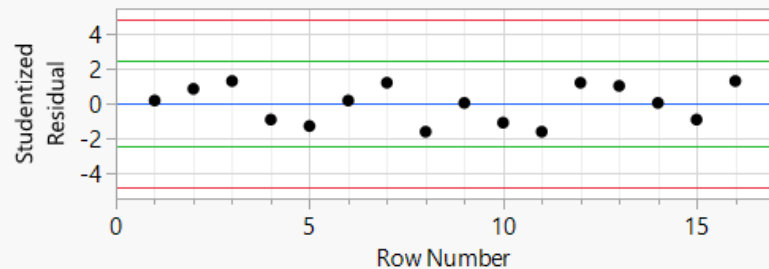


Second Model

Summary of Fit

RSquare	0.995851
RSquare Adj	0.991108
Root Mean Square Error	2.547758
Mean of Response	399.1875
Observations (or Sum Wgts)	16

Studentized Residuals



Externally studentized residuals with 95% simultaneous limits (Bonferroni) in individual limits in green.

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	8	10905.000	1363.13	210.0000
Error	7	45.438	6.49	Prob > F
C. Total	15	10950.438		<.0001*

Effect Tests

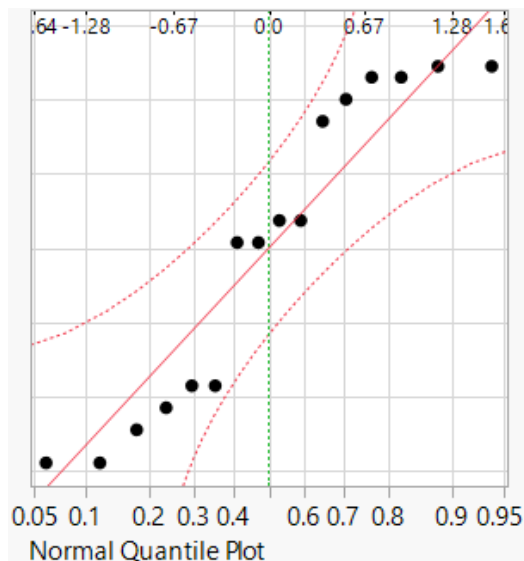
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temperature	1	1	7439.0625	1146.045	<.0001*
Time	1	1	1314.0625	202.4415	<.0001*
Pressure	1	1	430.5625	66.3315	<.0001*
GasFlow	1	1	10.5625	1.6272	0.2428
Temperature*Time	1	1	1139.0625	175.4814	<.0001*
Temperature*Pressure	1	1	451.5625	69.5667	<.0001*
Time*Pressure	1	1	60.0625	9.2531	0.0188*
Time*GasFlow	1	1	60.0625	9.2531	0.0188*

Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.868811	0.0261*
	A ²	Simulated p-Value
Anderson-Darling	0.7579157	0.0372*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

In my opinion the first model is better: **simpler** and assumptions **NOT rejected**



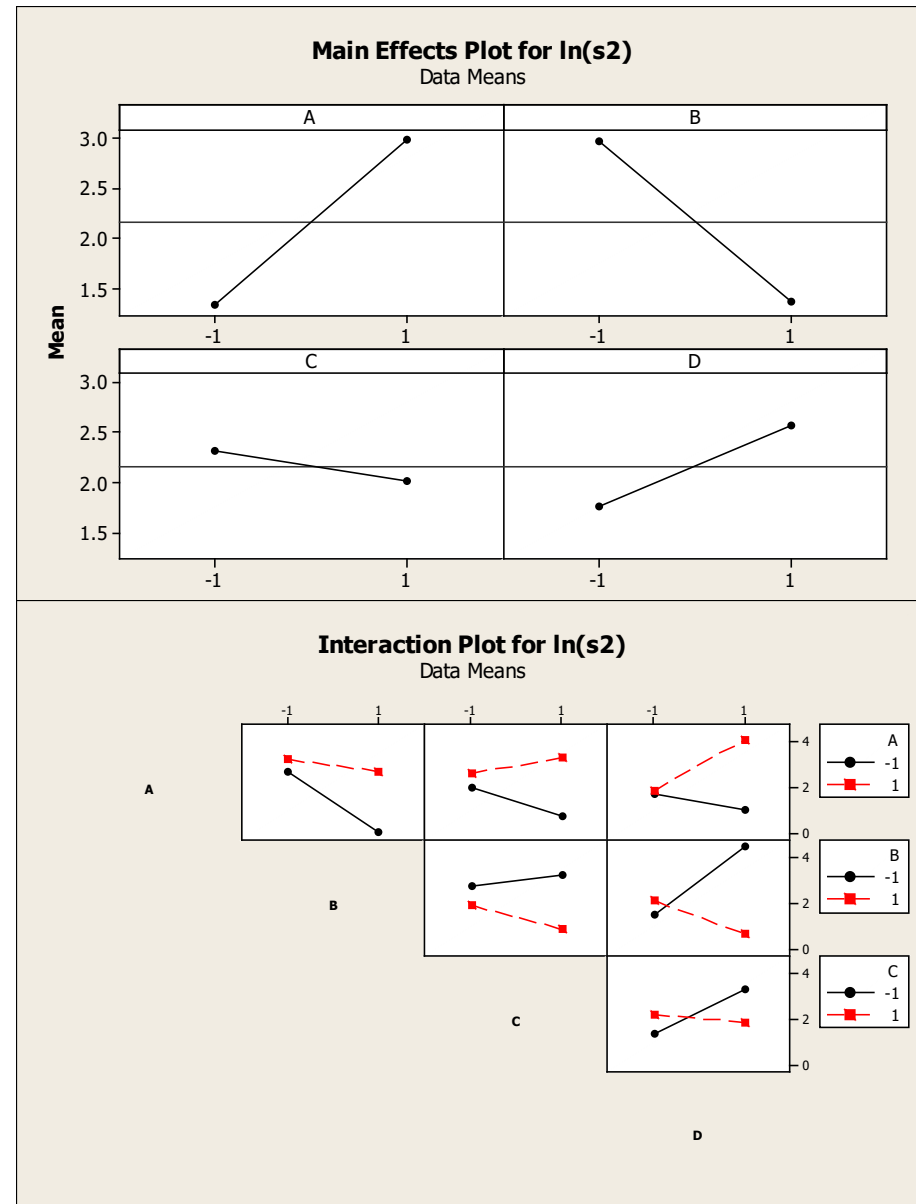
Dispersion Analysis

We analyse $\ln(s^2)$

The goal is to find out if there is a combination of levels that could reduce the dispersion

Possible Qualitative Model:
Temp, Time, Temp*Time

A Temperature
B Time
C Pressure
D Gas flow



Daniel Plot+Lenth (alfa=5%)

Effect Screening

The parameter estimates have equal variances.

The parameter estimates are not correlated.

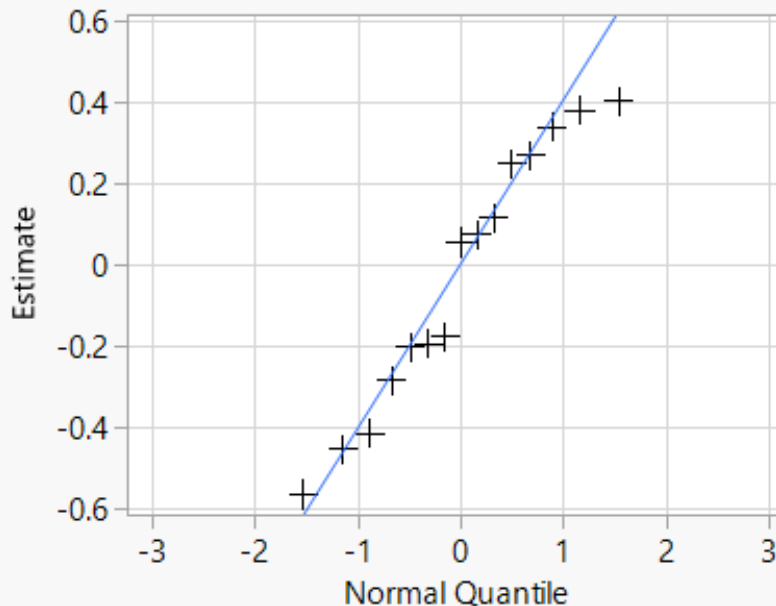
Lenth PSE

0.4025867

Orthog t Test used Pseudo Standard Error

Normal Plot

Normal Plot ▼



Summary of Fit

RSquare	0.277123
RSquare Adj	0.096404
Root Mean Square Error	1.183166
Mean of Response	1.081688
Observations (or Sum Wgts)	16

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temperature	1	1	2.7078529	1.9343	0.1895
Time	1	1	2.5795317	1.8427	0.1996
Temperature*Time	1	1	1.1525406	0.8233	0.3821

Conclusion: we don't have any evidence to reject the hypothesis that dispersion doesn't change with the factor levels but Montgomery concludes **differently** in the textbook, even if he remarks that the influence of the factors is not important

Central points in 2^k factorial plan

- The main idea is to replicate **only** the central point $(0,0,\dots,0)$ of the design.
- Central points give the **pure error variance** estimate and allow to distinguish between two possible models :

$$y = c_0 + \sum_{i=1}^k c_i x_i + \sum_{i=1}^k \sum_{j>i}^k c_{ij} x_i x_j + \varepsilon$$

$$y = c_0 + \sum_{i=1}^k c_i x_i + \sum_{i=1}^k \sum_{j>i}^k c_{ij} x_i x_j + \sum_{i=1}^k c_{ii} x_i^2 + \varepsilon$$

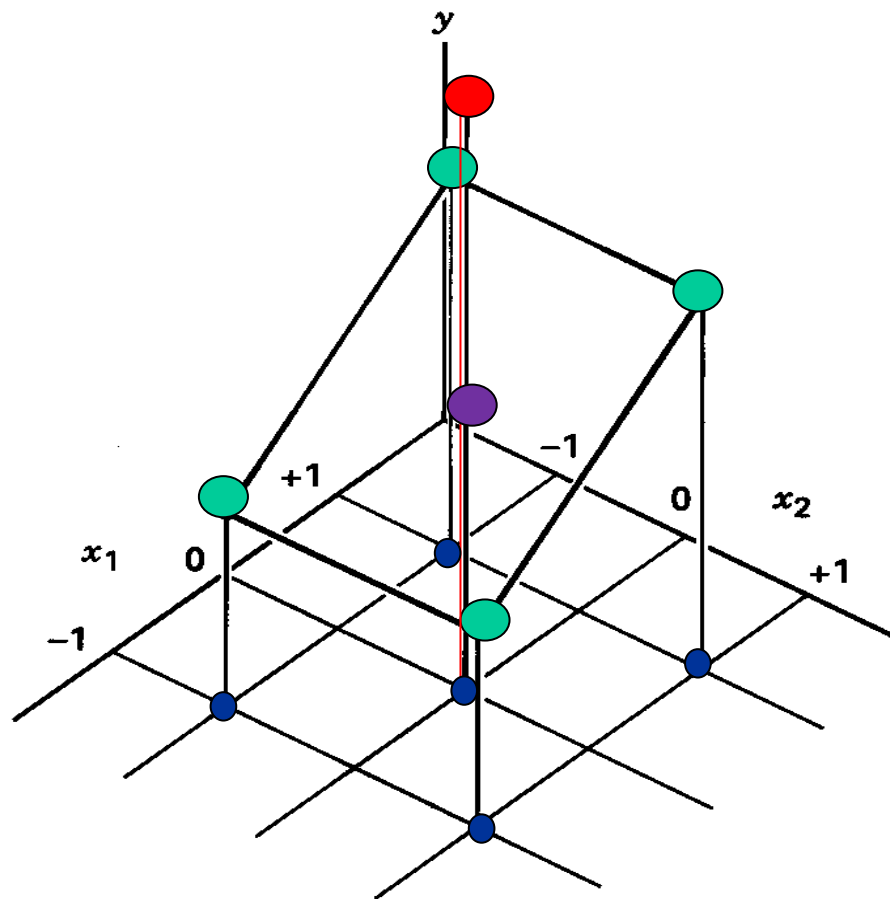


Figure 6-34 A 2^2 design with center points.

The **central points** can be used to estimate the **error variance** (Pure Error) with $(n_c - 1)$ degrees of freedom in unreplicated factorial plan and full model.

The **Curvature test** is:

$$SS_{\text{Curv}} = \frac{n_F n_C}{(n_F + n_C)} (\bar{y}_F - \bar{y}_C)^2 \quad df_{\text{Curv}} = 1$$

$$H_0 : c_{11} = c_{22} = \dots = c_{kk} = 0 \quad \text{vs} \quad H_0 : c_{11} + c_{22} + \dots + c_{kk} \neq 0$$

$$\frac{SS_{\text{Curv}} / 1}{MS_E} > F_{\alpha} (1, df_E)$$

or equivalently

$$\frac{n_F n_C}{(n_F + n_C)} |\bar{y}_F - \bar{y}_C| \frac{1}{MS_E} > t_{\alpha/2} (df_E)$$

Proof

$$E(\bar{y}_F) = \frac{1}{n_F} (n_F c_0 + n_F c_{11} + n_F c_{22} + \dots + n_F c_{kk}) = c_0 + c_{11} + c_{22} + \dots + c_{kk}$$

$$E(\bar{y}_F - \bar{y}_C) = c_{11} + c_{22} + \dots + c_{kk}$$

$$E(\bar{y}_C) = \frac{1}{n_C} (n_C c_0) = c_0$$

$$V(\bar{y}_F - \bar{y}_C) = \sigma^2 \left(\frac{1}{n_F} + \frac{1}{n_C} \right) = \frac{n_F + n_C}{n_F n_C} \sigma^2$$

then

$$t_0 = \frac{\bar{y}_F - \bar{y}_C}{\sqrt{\frac{n_F + n_C}{n_F n_C} MS_E}} \sim t(n_C - 1)$$

$$|t_0| > t_{\alpha/2}(n_C - 1) \quad \longrightarrow \quad \text{Reject } H_0$$

Example 6-6

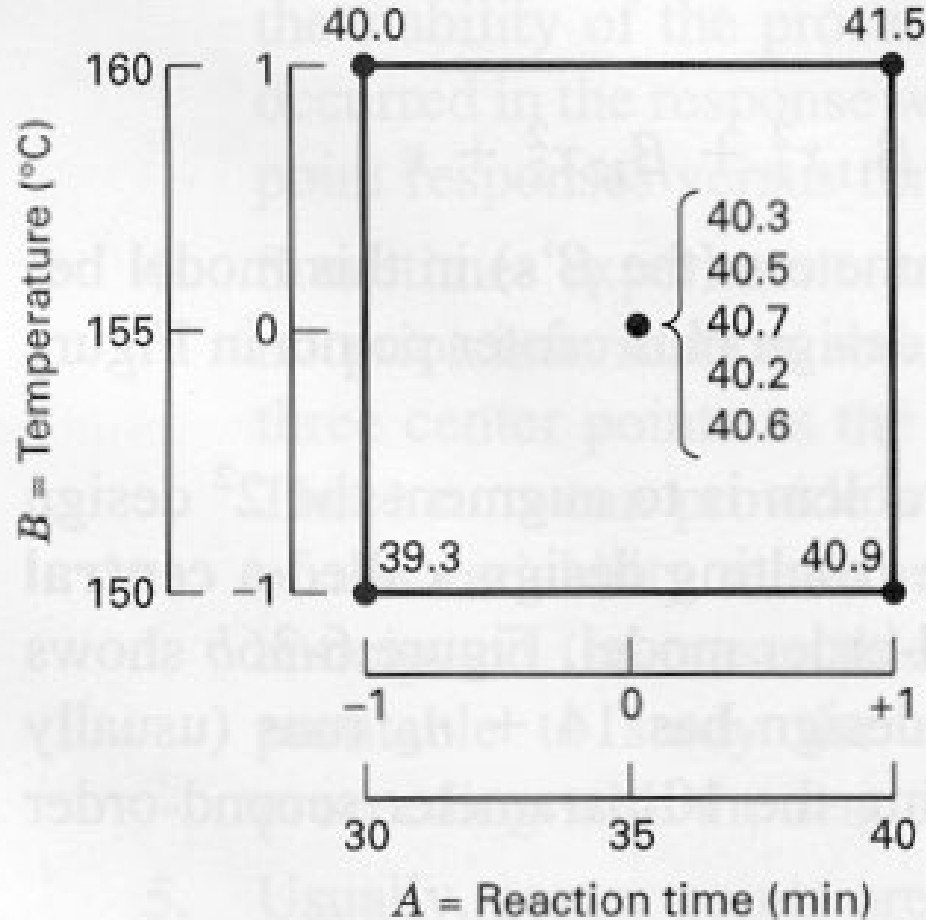


Figure 6-35 The 2^2 design with five center points for Example 6-6.

Response variable:
Process Yield

$$n_C = 5$$

Usually the number of central points is in the range (3-6)

JMP gives the pure quadratic curvature analysis using the F test

ANOVA table for Example 6-6

Summary of Fit

RSquare	0.942709
RSquare Adj	0.885418
Root Mean Square Error	0.207364
Mean of Response	40.44444
Observations (or Sum Wgts)	9

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	4	2.8302222	0.707556	16.4548
Error	4	0.1720000	0.043000	Prob > F
C. Total	8	3.0022222		0.0095*

Effect Tests

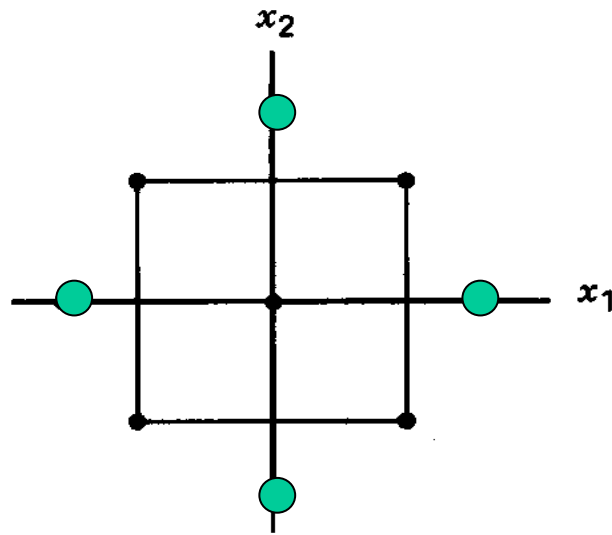
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
ReactionTime	1	1	2.4025000	55.8721	0.0017*
Temperature	1	1	0.4225000	9.8256	0.0350*
ReactionTime*Temperature	1	1	0.0025000	0.0581	0.8213
Temperature*Temperature	1	1	0.0027222	0.0633	0.8137

Curvature Term

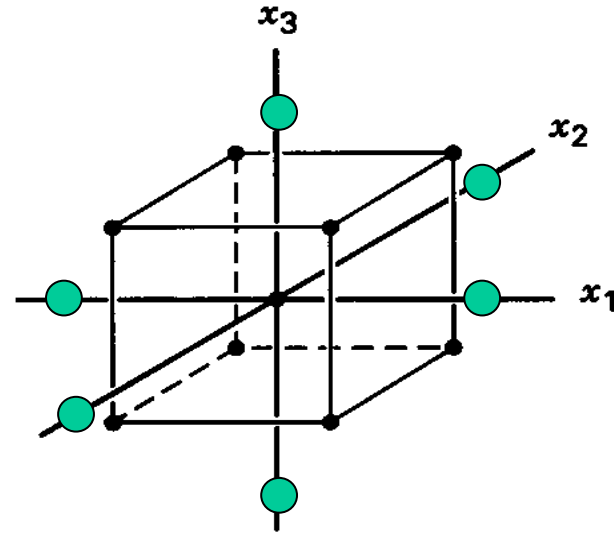
$$SS_E = \sum_{\text{Central Points}} (y_i - \bar{y}_C)^2 = 0.172 \quad df_E = 5 - 1 = 4 \quad MS_E = \frac{SS_E}{n_C - 1} = 0.043$$

$$\frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{(n_F + n_C) MS_E} = \frac{4 * 5 (40.425 - 40.46)^2}{(4 + 5) * 0.043} = 0.0633$$

If we find a significant **pure quadratic effect** it is necessary to enlarge the plan by adding some axial points, thus creating a CCD (**Central Composite Design**).



(a) Two factors



(b) Three factors

Figure 6-36 Central composite designs.

Central points for qualitative factors

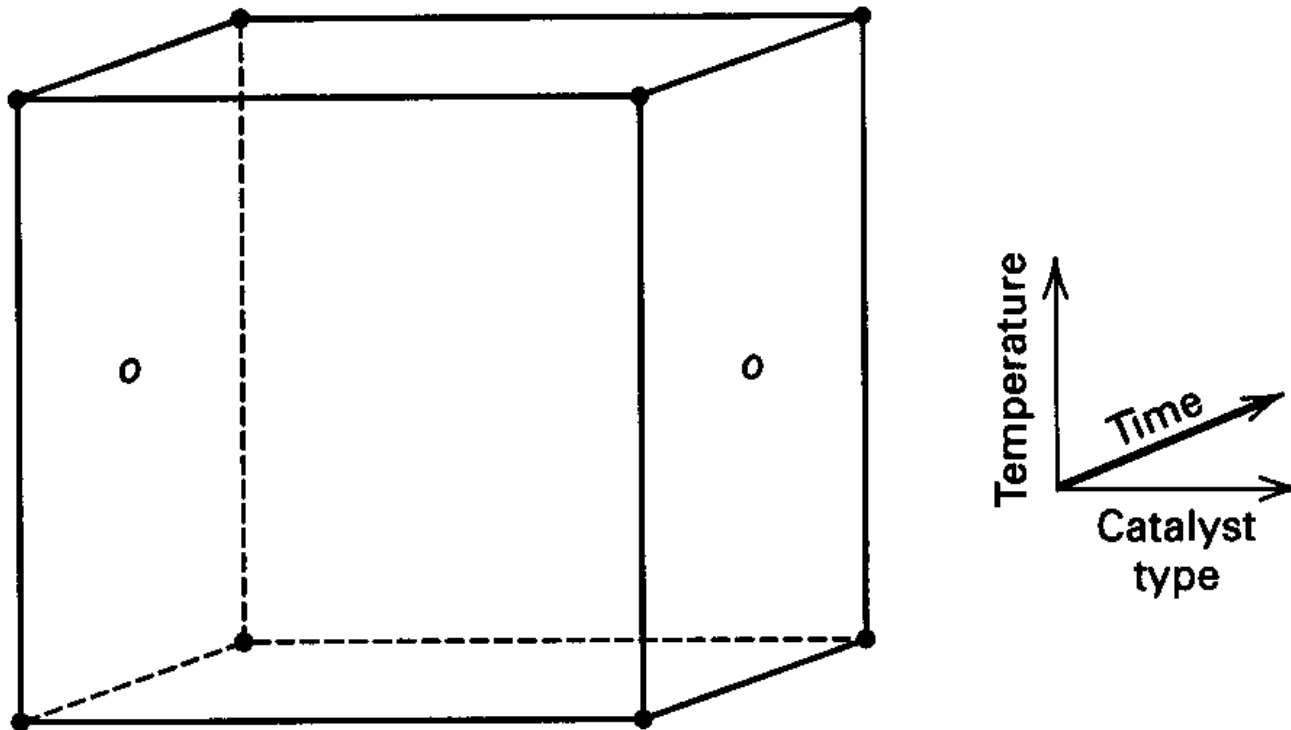


Figure 6-37 A 2^3 factorial design with one qualitative factor and center points.

Central points: summing up

- Use the **current process conditions** as replicated central points. This is useful to:
 - check a possible **time trend** in the response
 - check possible experimental set up **malfunctioning**
 - **start** the experiments using the central points (**but disrupting the randomization!**) to have an error variance estimate and to design properly the plan
- Pay attention to central points for **qualitative factors**

Power of a 2^k design

We have two main situations:

- Replicated design -> Use the standard method
- Not replicated design -> Compute the power of the reduced model

To compute the power of a **replicated** factorial 2^k we use the same methods, previously discussed for full & vbalanced designs.

$$\text{Power}_{[\]} = \Pr \left\{ F \left(df_{[\]}, df_E, \delta_{[\]} \right) > F_{\alpha} \left(df_{[\]}, df_E \right) \right\}$$

i.e. for A

$$\Delta_A^2 = \frac{n * b * c * .. * k}{df_A} \sum_{i=\{+,-\}} \tau_i^2 = \frac{n 2^{k-1}}{1} \frac{d_A^2}{2} = n 2^{(k-1)-1} d_A^2$$

$$\delta_A = \frac{\Delta_A^2}{\sigma^2} df_A = n 2^{(k-1)-1} \left(\frac{d_A}{\sigma} \right)^2 \quad df_A = 1$$

$$\Delta_{AB}^2 = \frac{n^* c^* d^* \dots^* k}{df_{AB}} \sum_{i,j} \tau \beta_{ij}^2 = n 2^{k-2} \frac{d_{AB}^2}{2} = n 2^{(k-2)-1} d_{AB}^2$$

$$\delta_{AB} = \frac{\Delta_{AB}^2}{\sigma^2} df_{AB} = n 2^{(k-2)-1} \left(\frac{d_{AB}}{\sigma} \right)^2 \quad df_{AB} = 1 \quad \leftarrow$$

$$\Delta_{AB..I}^2 = n 2^{(k-i)} \frac{d_{AB..I}^2}{2} = n 2^{(k-i)-1} d_{AB..I}^2 \quad df_{AB..I} = 1 \quad \leftarrow$$

$$\delta_{AB..I} = \frac{\Delta_{AB..I}^2}{\sigma^2} df_{AB..I} = n 2^{(k-i)-1} \left(\frac{d_{AB..I}}{\sigma} \right)^2$$

$$\Delta_{AB..I..K}^2 = n 2^{(k-k)} \frac{d_{AB..I}^2}{2} = n 2^{-1} d_{AB..I}^2 \quad df_{AB..I} = 1 \quad \leftarrow$$

$$\delta_{AB..I..K} = \frac{\Delta_{AB..I}^2}{\sigma^2} df_{AB..I} = n 2^{-1} \left(\frac{d_{AB..I}}{\sigma} \right)^2$$

Remarks on *df* degrees of freedom

$$df_{TOT} = N - 1 = (n2^k + n_c) - 1$$

$$df_{MOD}(\text{FULL}) = (2^k - 1) + 1(\text{Curvature})$$

$$df_{MOD}(\text{RED}) = (2^k - 1) + 1(\text{Curvature}) - n_{\text{Removed Terms}}(\text{including Curvature if eliminated})$$

$$df_E = 2^k (n - 1) + (n_c - 1) + n_{\text{Removed Terms}}(\text{including Curvature})$$

Example

Compute the power of the **6-6 experiment** (curvature test) to discover a difference not less than $d/\sigma=1$ on the **Main Effect** and on the **AB interaction**

$k=2$ $n_C=5$ **Full Model** with the **Curvature term**

Power and Sample Size (MTB confirmed by JMP)

2-Level Factorial Design

Alpha = **0.05** Assumed standard deviation = 1

Factors: **2** Base Design: 2; 4

Blocks: none

Including a term for center points in model.

Center		Total		
Points	Effect	Reps	Runs	Power
5	1	1	9	0.122249

Power 0.129964

(MTB)

Not including a term for the center points in the model

$$df_A = 1 \quad df_E = 2^k (n-1) + (n_c - 1) + n_{\text{Removed Terms}} = 0 + 4 + 0 = 4$$

$$\delta_A = n 2^{k-1-1} \left(\frac{d_A}{\sigma} \right)^2 = 1 \quad \text{including a term for the center points in the model}$$

$$\text{Power}_A = \text{Prob} \{ F(df_A, df_E, \delta_A) > F_\alpha(df_A, df_E) \}$$

Inverse Cumulative Distribution Function

F distribution with 1 df (numerator) and 4 df (denominator)

P(X <= x)	x
0.95	7.70865

Cumulative Distribution Function

F distribution with 1 df (numerator) and 4 df (denominator) and noncentrality parameter 1

x	P(X <= x)
7.70865	0.877751

$$\text{Power} = 1 - 0.877751 = \mathbf{0.122249}$$

- **Not** including the curvature term

$$df_A = 1 \quad \delta_A = 1$$

$$df_E = 2^k (n - 1) + (n_c - 1) + n_{\text{Removed Terms}} = 0 + 4 + \mathbf{1} = 5$$

Inverse Cumulative Distribution Function

F distribution with 1 df (numerator) and **5 df** (denominator)

$P(X \leq x)$	x
0.95	6.60789

Cumulative Distribution Function

F distribution with 1 df (numerator) and **5 df** (denominator) and noncentrality parameter 1

x	$P(X \leq x)$	
6.60789	0.870036	Power = $1 - 0.870036 = \mathbf{0.129964}$

- **Power for interaction AB including the curvature term**

$$df_{AB}=1 \quad \delta_{AB} = n2^{k-2-1} \left(\frac{d_{AB}}{\sigma} \right)^2 = 0.5$$

$$df_E = 2^k (n-1) + (n_c - 1) + n_{\text{Removed Terms}} = 0 + 4 + 0 = 4$$

Inverse Cumulative Distribution Function

F distribution with 1 *df* (numerator) and 4 *df* (denominator)

P(X <= x)	x
0.95	7.70865

Cumulative Distribution Function

F distribution with 1 *df* (numerator) and 4 *df* (denominator) and **noncentrality parameter 0.5**

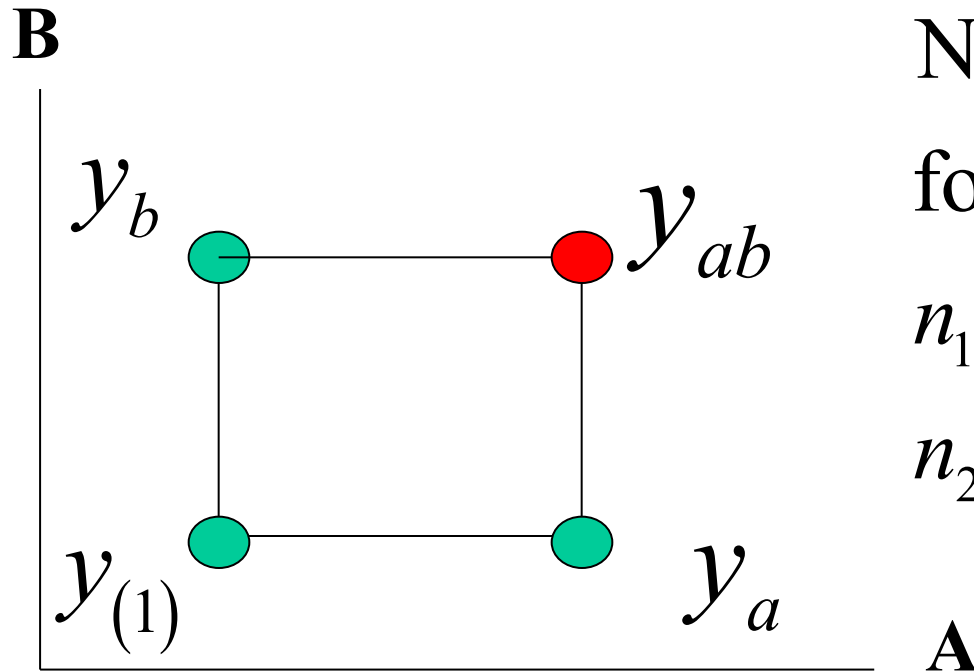
x	P(X <= x)
7.70865	0.914122

$$\text{Power} = 1 - 0.914122 = \mathbf{0.085878}$$

- **How many replicates are necessary to have a 90% power?**
- **Skip the paragraph 6.7**

- We have shown that a **Factorial Design** is more effective if the factors have interactions (**Efficacy**)
- Now we will show that the **Factorial Design** is more **Efficient** than OFAT
- We measure the **Efficiency** with the **Estimate precision** (i.e. the Variance) of the model parameters.
- If we have a **two factors design**, the **Efficiency** is **9/8** compared with a OFAT design.
- **Efficiency increases** as the number of factors increases

Example with two factors A and B



Number of replications
for each treatment

n_1 OFAT $\rightarrow 1$

n_2 Factorial $\rightarrow 2$

$$EFF_{B,1} = \frac{y_b}{n_1} - \frac{y_{(1)} + y_a}{2n_1} \rightarrow V(EFF_{B,1}) = \frac{\sigma^2}{n_1} + \frac{\sigma^2}{2n_1} = \frac{3\sigma^2}{2n_1}$$

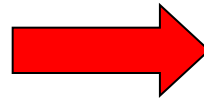
$$EFF_{B,2} = \frac{y_b + y_{ab}}{2n_2} - \frac{y_{(1)} + y_a}{2n_2} \rightarrow V(EFF_{B,2}) = \frac{\sigma^2}{2n_2} + \frac{\sigma^2}{2n_2} = \frac{\sigma^2}{n_2}$$

$$EFF_{B,1} = \frac{y_b}{n_1} - \frac{y_{(1)} + y_a}{2n_1} \rightarrow V(EFF_{B,1}) = \frac{\sigma^2}{n_1} + \frac{\sigma^2}{2n_1} = \frac{3\sigma^2}{2n_1}$$

$$EFF_{B,2} = \frac{y_b + y_{ab}}{2n_2} - \frac{y_{(1)} + y_a}{2n_2} \rightarrow V(EFF_{B,2}) = \frac{\sigma^2}{2n_2} + \frac{\sigma^2}{2n_2} = \frac{\sigma^2}{n_2}$$

$$\frac{V(EFF_{B,1})}{V(EFF_{B,2})} = \frac{3}{2} \frac{n_2}{n_1} \quad \rightarrow \quad N = 3n_1 = 4n_2 \rightarrow \frac{n_2}{n_1} = \frac{3}{4}$$

If we have the same experimental effort **N**



$$\frac{V(EFF_{B,1})}{V(EFF_{B,2})} = \frac{9}{8}$$